When Does Training on Downscaled Images Yield the Same Gradients?

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Abstract

Diffusion transformers deliver strong image generation, but their training cost grows superlinearly with resolution. A recent family of methods therefore trains or samples at reduced resolution on a spectral premise: at high noise, a downscaled latent preserves almost the full surviving signal. The evidence for that premise, however, has been gathered at the level of generated output, where the substitution is observed only through the confounds of a full pipeline; whether a downscaled step also delivers the *native training gradient direction*, under the same model and objective, has remained unresolved. We propose a two-term account of what the substitution costs: a noise-dependent term governed by the downscale *ratio*, and a σ-independent floor governed by the target grid's *absolute token count* and carried by the compute graph itself. On the resulting (route, σ) map, the mildest route (1024→896) is measured safe against a re-encode control over the interior high-noise window, and the account uncovers a window that no spectral tolerance predicts: on the deeper 1024→768 route at $0.65 < \sigma < 0.95$, the downscaled step's gradient direction retains only a small residual excess over the native one (+0.02–0.04 per noise bin, certified at a relaxed margin of 0.09, short of strict certification at 0.02) atop a floor that no noise level removes. Gating substitution on the measured-safe route yields *selective* low-resolution LoRA training with a 14.6% smaller training footprint at a fixed step budget, while producing near-native adapter weights. With demotion scheduled late and both routes stacked, the trained ΔW matches its matched-seed native twin at cosine 0.75, well above the 0.26 at which an identical retrain reproduces itself once deterministic kernels are disabled; a change of training seed alone reads 0.121, and demoting 1024→896 at every σ reads 0.183.

1 Introduction

The cost of a diffusion-transformer forward pass grows superlinearly with resolution: token count grows quadratically with edge length, and attention cost grows again in token count. This cost structure makes the title question worth stating precisely. At high noise, can the *gradient* of an adapter be computed on a downscaled latent as a drop-in substitute for the native one, with the same model, the same objective, and the same σ distribution, and with only the spatial grid changed on some steps?

There is a well-studied reason to expect an affirmative answer. As the noise level σ grows, per-frequency signal-to-noise falls, and high spatial frequencies drop below the noise floor first, so that above a spectral crossover a downscaled latent carries *almost* the full surviving signal. Studies of this coupling, however, are confined almost entirely to the *inference* side: scale-wise distillation (Starodubcev et al., 2025), spectral progressive diffusion (Xiao et al., 2026), and spectrally-guided noise schedules (Esteves & Makadia, 2026) run high-noise *sampling* steps at reduced resolution and judge the premise by the generated output. Where resolution is reduced during training (Jin et al., 2024; Karras et al., 2018; Chen et al., 2024; Hoogeboom et al., 2023), the low-resolution stage is given its own, modified objective, and success is again judged from the final samples. By the spectral premise, substitution is safe *whenever the noise has masked the frequencies the coarse grid loses*.

None of this evidence resolves the question at *training* time: whether a downscaled step delivers the native gradient direction under an unchanged objective. Output-level success observes the substitution only through the confounds of a full training or sampling pipeline, and a small difference in a quality score does not imply a small difference in what a training step *learns*. Measuring that gap directly is the route this paper takes. What a training step consumes is the expected adapter gradient, and *demotion*, our term for the training-time substitution of a downscaled, re-encoded latent for the native one, perturbs more than the network's input: the regression target carries the clean image at unit weight at every noise level, and the compute graph itself changes with the grid. Measuring the gap honestly turns out to be a metrology problem. The natural estimator, a redraw floor minus a single demoted-arm cosine, is biased by finite-draw variance, and the bias grows as the demoted grid shrinks, manufacturing exactly the token-count-ordered floor signature under test. Every claim below is therefore stated in debiased units, at the instrument's measured resolution.

The contributions:

1. **Two accounts, stated in the same units** (§3). We define the demotion gap at the gradient level, then derive in those units (a) the gradient gap implied by the existing spectral argument, taken in its strongest tolerance-parameterized form (SPD, Xiao et al., 2026), which is *ratio-only* and *floorless* by construction, and (b) our own account, which treats demotion as a perturbation of both factors of $g = J^\top r$ and reduces, under four named assumptions, to a route amplitude on a universal mismatch curve *plus* a σ-independent graph floor. The two differ structurally before any measurement (§3.2), and every experiment is a point of disagreement between them.
2. **A debiased measurement that scores both, and a predictive tool** (§4; the resulting (route, σ) map in §4.2). Measured with per-arm self-floors, a draw-count extrapolation, and a pre-specified margin whose certification power is set, though not defined, by the instrument's resolution, the map reads as follows: the mildest route (1024→896) is certified non-inferior to the re-encode control over the interior high-noise window $\sigma \in (0.5, 0.94]$; the deeper 1024→768 route retains a narrow low-excess window ($0.65 < \sigma < 0.95$) atop a persistent floor; and no noise level rescues a 2× downscale. No member of the spectral tolerance family reproduces this map; in particular, none locates the 1024→768 window, nor the absence of any window for 1024→512. Head to head on the same curves, the spectral family misses at 0.147–0.355 RMSE, whereas our account predicts held-out routes at ~0.07–0.09, better than an oracle fit on the held-out data itself, and the floor law correctly predicts both held-out floors.
3. **Selective low-resolution training** (§5). Gating substitution on the measured-safe window gives an opt-in σ-conditional trainer realizing −15.1% forward FLOPs and −14.6% wall-clock at a fixed step budget, with a weight-space endpoint footprint a factor of 2–3 inside the training-seed lottery, rising to endpoint cosine 0.75 when demotion is late-scheduled (Table 1). Generation-quality non-inferiority at production scale remains the stated open gate.

2 Related work

Flow matching and DiT. Flow matching (Lipman et al., 2023) regresses a velocity field along prescribed probability paths between noise and data; rectified flow (Liu et al., 2023) takes the paths to be straight lines, giving the linear noising $z_\sigma = (1-\sigma)x + \sigma\epsilon$ and the constant target $v = \epsilon - x$ used throughout, the training objective of current large text-to-image systems (Esser et al., 2024). The Diffusion Transformer (Peebles & Xie, 2023) replaces the U-Net denoiser with a transformer over patchified latent tokens, so a spatial grid enters the network only as a token sequence: token count grows quadratically with edge length and attention cost quadratically again in token count, the cost structure that makes resolution the expensive axis. Spatial position is now widely carried by a grid-dependent coordinate system, most commonly rotary embeddings (Su et al., 2024).

Scale-wise and progressive-resolution diffusion. Progressive growing (Karras et al., 2018) and resolution curricula (Chen et al., 2024; Hoogeboom et al., 2023) train at increasing resolution as a schedule, each low-resolution stage carrying its own stage-specific objective. A more recent family ties resolution to the noise level itself, on the spectral premise above, and has so far been studied on the *inference* side. Scale-wise distillation (SwD) (Starodubcev et al., 2025) states the claim directly (above the crossover, the noised downscaled latent is close in distribution to the downscaled noised latent) and samples early steps at reduced resolution. Spectral Progressive Diffusion (SPD) (Xiao et al., 2026) gives the argument its sharpest available form: a tolerance-parameterized per-frequency activation time under a diagonal Gaussian spectral model, a criterion on the per-band output of the *Bayes-optimal predictor* rather than merely on input SNR, from which an inference-time resolution schedule follows. Spectrally-guided noise schedules (Esteves & Makadia, 2026) apply the same reasoning to schedule design; where the premise does enter training, as in pyramidal flow matching (Jin et al., 2024), the objective is restaged per pyramid level rather than left native.

Positional extension. Position interpolation (Chen et al., 2023), banded/frequency-selective variants (YaRN, Peng et al., 2024), and diffusion-specific dynamic position extrapolation (Issachar et al., 2026; Zhao et al., 2026) rescale rotary coordinates to evaluate a model at unseen grid sizes. We use exact positional interpolation as a *causal intervention* that matches the downscaled grid's relative phase geometry to native, in order to decompose the resolution-sensitivity floor, and adopt the σ-gated band schedule of Zhao et al. (2026) as a training-time refinement.

3 Two accounts of what demotion costs

This section fixes the estimand (§3.1), then states two accounts of it in the same units: a training-gap interpretation built on the spectral account of the inference-side literature (§3.2), and our own (§3.3). The two are not variants of one model: they disagree about which quantity governs a route's cost, about what happens at pure noise, and about whether a single parameter can order all routes.

3.1 The estimand

Objective and noising. A flow-matching model regresses the constant-velocity transport between a clean latent and pure noise. With clean latent x, noise $\epsilon \sim \mathcal{N}(0, I)$, and noise level $\sigma \in (0, 1]$, the noised input is $z_\sigma = (1 - \sigma)\, x + \sigma\, \epsilon$ and the target is the velocity $v = \epsilon - x$; with caption condition c the loss is $\|\hat{v}_\theta(z_\sigma, \sigma, c) - v\|^2$.

Demotion. We write $e_0 \to e$ for a *demotion*: the native latent at the route's source edge e_0 is replaced with a coarser-grid one at its target edge e. With y the image at its native resolution, $\mathcal{E}$ the VAE encoder, and $\mathcal{R}_e$ the pixel-space downscale to the tier-e bucket (native aspect ratio preserved), $x_{\mathrm{src}} = \mathcal{E}(y) \longrightarrow x_{\mathrm{dem}} = \mathcal{E}\big(\mathcal{R}_e(y)\big)$, yielding a latent with proportionally fewer spatial tokens; the downscale acts in pixel space, not latent space. The demoted step trains on $z_\sigma = (1 - \sigma)\, x_{\mathrm{dem}} + \sigma\, \epsilon$ with target $v = \epsilon - x_{\mathrm{dem}}$, so the substitution changes both what the network sees and what it is asked to predict, but only through the spatial grid.

The estimand. Three *arms* recur throughout: the source arm src, which trains on the native-resolution latent x_{src}; the demoted arm dem, which trains on x_{dem}; and the control arm reenc, which decodes and re-encodes the native latent at native resolution. For an arm $a \in \{\mathrm{src}, \mathrm{reenc}, \mathrm{dem}\}$, let $\bar{g}_a(\sigma) = \mathbb{E}_\epsilon[\nabla_\theta \mathcal{L}]$ be the population per-bin mean adapter gradient of a query (image, caption). The *population demotion distance* of the route, $d_{e_0 \to e}$, is

$$d_{e_0 \to e}(\sigma) \;=\; 1 - \cos\big(\bar{g}_{\mathrm{src}}(\sigma),\, \bar{g}_{\mathrm{dem}}(\sigma)\big) \;=\; \tfrac{1}{2}\left\|\hat{g}_{\mathrm{src}} - \hat{g}_{\mathrm{dem}}\right\|^2 \;\geq\; 0, \tag{1}$$

with $\hat{g}$ the unit-normalized gradients. Route-level scalars carry the *route* as a subscript, abbreviated to the target edge alone (d_e) wherever the source is fixed by context; the control's distance d_{reenc} carries its arm label instead, since it changes no grid (full convention: Appendix A.1). The *reported* quantity of every bin-resolved map and every fit in this paper is the *re-encoding excess*

$$\mathrm{gap}_e(\sigma) \;:=\; d_e(\sigma) \;-\; d_{\mathrm{reenc}}(\sigma), \tag{2}$$

the demotion distance beyond what the control already pays; gap_e can be negative, d_e cannot. Endpoint and floor tables report debiased estimates of d_e with the control as its own row. Every map in the body aggregates per example, taking the cosine per image and then the mean; the batch-aggregate object that SGD follows in batched training is a distinct estimand, treated in Appendix A.1, and where measured it reads as the *more forgiving* of the two (§4.2).

3.2 The spectral account

The spectral argument of §2 supplies the field's working model of the same gap, and it is worth deriving in the units Eq. 1 just fixed. Its strongest available form is the tolerance-parameterized crossover of Xiao et al. (2026).

Demotion deletes a band. Write $u^{(\omega)}$ for the frequency-ω component of a field u in the latent's spatial Fourier decomposition, with ω measured in cycles per sample of the *native* grid, so that the native Nyquist frequency sits at $\frac{1}{2}$. The downscale $\mathcal{R}_e$ resamples the image with e/e_0 times fewer samples per axis, and a grid at that density can represent no frequency above half its own sampling rate; the resampler's anti-aliasing prefilter removes, rather than aliases, whatever lies above. Idealizing the VAE round trip away, whose residue is exactly what the reenc control of §3.1 pays, demotion is an ideal low-pass:

$$x_{\text{dem}}^{(\omega)} = \begin{cases} x^{(\omega)}, & |\omega| < \omega_e, \\ 0, & |\omega| \geq \omega_e, \end{cases} \qquad \omega_e = \frac{1}{2}\frac{e}{e_0}, \tag{3}$$

deleting exactly the bands above the coarse grid's Nyquist frequency ω_e and leaving the rest intact.

Carried to the training gradient. The noising is linear, so it acts band by band, and under the premise that each clean-latent band is independently Gaussian, $x^{(\omega)} \sim \mathcal{N}(0, P_\omega)$ with P_ω the band's clean-latent power, Xiao et al. (2026) attach to each band an activation time: a crossover t_ω past which the Bayes-optimal per-band velocity prediction sits within a tolerance δ of the pure-noise answer $\epsilon^{(\omega)}$. The tolerance δ is the account's one free parameter; its instantiations are explored in Appendix D, and no conclusion below hinges on its choice. The destroyed bands of Eq. 3 all sit at or above ω_e; on a decaying spectrum the lowest of them is the most powerful, so it crosses last, and the safe-substitution boundary for the whole route sits at t_{ω_e}. Under the premise that demotion costs only what the destroyed bands still contribute to the prediction, the cost has one place to land: whatever those bands contribute to the residual, and through the backward pass to the adapter gradient, perturbs $\bar{g}_{\text{dem}}$ away from $\bar{g}_{\text{src}}$. The implied gradient gap is that contribution, gated at the Nyquist band's activation time:

$$\text{gap}_e^{\text{spec}}(\sigma) = S_e(\sigma)\,\mathbf{1}\big[\sigma < t_{\omega_e}\big], \qquad t_\omega = \frac{1}{1 + \sqrt{\delta \,/\, \big(P_\omega\,(1 + P_\omega - \delta)\big)}}, \tag{4}$$

where $S_e(\sigma)$ is the destroyed-band contribution to the distance of Eq. 1. S_e can be computed. At the residual level the premise is complete: the Gaussian posterior mean is the classical per-band Wiener shrinkage (Wiener, 1949), so the cross-grid mean-residual mismatch is confined to the destroyed band, ordered across routes by destroyed-band energy, and follows in closed form from the power spectrum (Appendix A.3). Carrying it into the gradient units of Eq. 1 costs one calibrated gain, since the map passes through the network's Jacobian, about which a spectral model of the *data* says nothing. Appendix D instantiates exactly this pipeline when the account is scored.

Our claim is therefore not that S_e cannot be estimated, for it can be, one way or another, but that a gap estimate built from it is structurally insufficient for gap_e. Three limits are fixed by the *form* of Eq. 4, before any gradient is measured and however S_e is computed, and each is a point at which the account can be falsified without arguing about δ. **(a) One parameter orders every route**: all boundaries move together under the single tolerance, so the family is totally ordered. **(b) Ratio only**: the cut $\omega_e = \frac{1}{2}\, e/e_0$ of Eq. 3 is a function of the downscale *ratio* alone, so the estimate is structurally blind to any absolute-size effect. **(c) No floor**: for every route and every δ the predicted gap vanishes above a finite boundary, so no σ-independent term is expressible.

3.3 Our account: what demotion perturbs, and its angular reduction

We start one level lower, from what the substitution touches in the gradient itself. Differentiating the loss of §3.1 gives, per draw and up to a constant, $g = \nabla_\theta \frac{1}{2}\|\hat{v}_\theta - v\|^2 = J^\top r$, with $r = \hat{v}_\theta - v$ the prediction residual and $J = \partial \hat{v}_\theta / \partial\theta$ the Jacobian through which the compute graph turns a residual into an adapter gradient. Demotion perturbs both factors: the residual, through the data the objective consumes, and the Jacobian, through the compute graph. The estimand is built from noise means, so the relevant residual object is the *mean residual* at fixed query, $\bar{r}_a(\sigma) = \mathbb{E}_\epsilon[\hat{v}_\theta - v]$, under the same averaging that defines $\bar{g}_a$ in §3.1: a velocity-shaped field with a definite σ-curve for each arm. Write $\Delta\bar{r}(\sigma)$ for its cross-arm mismatch $\bar{r}_{\text{dem}} - \bar{r}_{\text{src}}$, and ΔJ likewise for the noise-mean Jacobians. Averaging each factor over draws and expanding the product (Appendix A.2), the perturbation $\delta g = \bar{g}_{\text{dem}} - \bar{g}_{\text{src}}$ decomposes as

$$\delta g \;=\; \underbrace{J^{\top}\Delta\bar{r}(\sigma)}_{\text{data branch } B_e} \;+\; \underbrace{\Delta J^{\top}\,\bar{r}}_{\text{graph branch } C_e} \;+\; \underbrace{\Delta J^{\top}\Delta\bar{r} + \cdots}_{\text{remainder } R_e}. \tag{5}$$

J_{src} and J_{dem} act on different grids, so the branches are not literal matrix products across arms; they are defined *operationally*, as the two independent perturbation directions in the shared adapter-parameter space where both gradients live, and §4.3 isolates each by its own intervention.

Read against Eq. 5, the spectral account of §3.2 keeps only the input-mediated part of the data branch, gated at Nyquist, and sets the rest to zero. The three limits fixed there are exactly the discarded terms read back: the graph branch is a property of the two grids, referencing neither σ (limit c) nor their ratio (limit b), and a two-term gap admits no single-parameter ordering (limit a). One discarded term matters later: the regression target carries the clean image at unit weight at every σ, so even at σ=1 the mean residual retains an image-dependent survivor, $x - \mathbb{E}[x \mid c]$ (Appendix A.3); whether it is resolvable in the angular estimand is an empirical question, discharged in §4.3.

The angular reduction. It remains to read Eq. 5 in the units of the estimand. The geometry is pure bookkeeping, so the displayed forms, together with the κ notation in which they are stated, are deferred to Appendix A.4; what the argument needs is their content. Splitting the perturbation δg into a rescaling of $\bar{g}_{\text{src}}$ and a rotation away from it, the gap is, *exactly*, a saturating function of a single rotation-to-scale ratio (Eq. 18): only the orthogonal component moves the estimand, and the cosine is *blind* to a parallel rescaling, a fact on which the endpoint probes of §4.3 rely. Expanding at small perturbation and substituting Eq. 5 gives the *four-term angular expansion* (Eq. 20), derived as unconditionally as the exact form itself. Writing $u^{\perp}$ for the component of u orthogonal to $\bar{g}_{\text{src}}$, its terms are a data share $S_e = \|B_e^{\perp}\|^2/(2\|\bar{g}_{\text{src}}\|^2)$, a graph share built likewise from $C_e^{\perp}$, both non-negative; a projected interaction I_e that can carry either sign, a fact that matters in Appendix C; and a remainder R'_e. Where the measured perturbation turns out large (the 512 route; §4.3), quadratic-order reads are indicative only and the exact form carries the quantitative claims. Note that Eq. 4 is Eq. 20 with the graph share and interaction zeroed and S_e gated at Nyquist.

From the expansion to a measurable form. The reported object is the excess over the control, whose own expansion is degenerate: the reenc arm changes no grid, so $\Delta J_{\text{reenc}} = 0$ and both terms built from it vanish, leaving $d_{\text{reenc}} \approx S_{\text{reenc}} + R'_{\text{reenc}}$. Subtracting term by term therefore cancels only the re-encode component of the data share and passes the graph share and interaction through untouched. Four named assumptions, each tested by a designated probe in §4, then trade the exact bookkeeping (in which no term is separately measurable per bin) for a form built from measurable curves and per-route constants: (i) **small remainder**, R'_e negligible over the claimed domain; (ii) **negligible projected interaction**, $|I_e|$ small against the retained shares; (iii) **graph-relative stationarity**, $\|C_e^{\perp}\|/\|\bar{g}_{\text{src}}\|$ σ-constant, since numerator and denominator share the residual's scale, so that the graph share is flat in σ: the per-route *floor* of Eq. 6; (iv) **data-branch factorization**, $\|B_e^{\perp}\| \approx a_e\,\|\Delta\bar{r}(\sigma)\|$: a σ-independent route amplitude times the *norm* of the cross-grid mean-prediction-residual mismatch of Eq. 5, asserted to be route-independent and directly measurable as such (§4.3); the measured curve is an *empirical* closure of the mean residual's exact posterior form (Appendix A.3), not an unconstrained fitted shape. The first two assumptions are not unstructured hopes: both discarded terms obey exact Cauchy–Schwarz bounds in the retained shares, the interaction by their geometric mean and the remainder entering at half power in its own share, so (ii) carries independent content only near share crossover (Appendix A.4). Applied,

$$\mathrm{gap}_e(\sigma) \;\approx\; \underbrace{\tfrac{1}{2}\,a_e^2 \cdot \big(\|\Delta\bar{r}(\sigma)\| \,/\, \|\bar{g}_{\text{src}}(\sigma)\|\big)^2}_{\text{data term}} \;+\; \underbrace{\mathrm{RoPE}_e + \mathrm{Resid}_e}_{\text{graph term (the floor)}}, \tag{6}$$

the *two-term account*, stated on the reported excess. The quadratic power is fixed by the cosine geometry *locally*; far from the small-mismatch regime the licensed read is the un-expanded parent form, Eq. 18 with loading $\kappa_e(\sigma) = a_e\,\|\Delta\bar{r}(\sigma)\|/\|\bar{g}_{\text{src}}(\sigma)\|$ (the same amplitude, recovering Eq. 6's data term in the small-κ limit), whose data term *saturates* instead of overshooting. Appendix C scores both forms held out: Eq. 18 is the best empirical predictor, Eq. 6 the derived small-perturbation form. The ledger is as follows: Eqs. 18 and 20 are unconditional; Eq. 6 is conditional on the four assumptions; and the coefficients a_e, $\|\Delta\bar{r}(\sigma)\|$, $\|\bar{g}_{\mathrm{src}}(\sigma)\|$, and each route's floor are measured. The account was pre-registered before the discriminating runs, with an outcome matrix that included a kill switch on the graph term (all endpoint gaps ≈ 0).

The graph term has identifiable parts. Changing the grid changes (i) the rotary coordinate system, in which every band of the position embedding accumulates phase at a different density across the image, and (ii) non-positional graph statistics: attention softmax over a different token count, sequence-length-dependent normalization, and the capacity of the coarse graph to approximate the fine graph's computation. The conceptual frame for the latter is attention as a discretization of a function-space operator (Calvello et al., 2024), whose statistics need not be discretization-invariant. These are the two parts RoPE_e and Resid_e written into Eq. 6. The split yields one sharp prediction: the positional share is a *coordinate-system artifact*, so an exact positional-interpolation intervention should remove RoPE_e wherever the data branch is inactive ($\sigma \to 1$); a band-counting sketch attaches an absolute-*size* governor to Resid_e. Like YaRN's own band thresholds (Peng et al., 2024), that sketch's constants are calibrated, not derived; we claim no closed form for the floor.

4 Experiments: both accounts, scored

This section scores both accounts of §3 against one measured object: debiased demotion-gap curves over the full σ axis, on routes neither account saw. We build the instrument (§4.1), read the per-route map its curves certify, which is the deployed artifact (§4.2), and then score both accounts against the same curves: the spectral account at the boundary and the curve level, ours term by term (§4.3). The held-out prediction protocol, its pre-registered gates, and the structural failures it exposes are quantified in Appendix C. Each subsection states the prediction it discharges against criteria frozen before the runs; several probes double as falsifiers of our own account's assumptions.

4.1 The instrument

Model and data. All measurements use Anima (CircleStone Labs & Comfy Org, 2025), an open DiT-based flow-matching text-to-image model (2B parameters, 28 transformer blocks, 3D rotary position embeddings, and the Qwen-Image VAE, Wu et al., 2025), fine-tuned with LoRA adapters on an illustration corpus. Images are bucketed by native aspect ratio into resolution *tiers* indexed by nominal edge length $e \in \{512, 768, 896, 1024, 1280\}$; a tier-$e$ image occupies roughly $(e/16)^2$ latent-patch tokens (~4,100 at 1024, ~3,000 at 896, ~2,160 at 768, ~1,000 at 512). The probe adapter is a plain LoRA checkpoint trained at native tiers; we return to this operating-point dependence in §6.

The gradient probe. For each image and each of B σ-bins we accumulate adapter gradients over D stratified noise draws per arm (verdict run: 14 interior bins on a segmented grid dense below $\sigma = 0.1$ and above 0.9, plus a σ=1 endpoint bin, D=12, deterministic kernels, $N = 40$ images) and compare arms by cosine similarity of the flattened accumulated gradients. The arms are: a **floor**, two independent draw sets at native resolution, the redraw null; the **reenc** control of §3.1; one **demote** arm per edge e; and, on debiased runs, per-arm **self-floors**, a second independent draw set for every arm. The σ=1 endpoint bin is a distinct probe *mode*: its input is pure ϵ, so the input-mediated data term vanishes by construction. An **x-zero** companion mode removes the image from input *and* target entirely. Both modes read only at the endpoint, since with the $(1-\sigma)x$ term absent the lower bins are off-manifold (Appendix B), and together they carry the floor measurement of §4.3. The gap is in cosine units, a scale-free mismatch *fraction*, with bin-mean SEM 0.01–0.07 at N=40. Gap subtraction, not absolute cosines, is the valid read: the floor itself moves with $\|g\|$ across bins. Every bin-mean curve must pass a split-half reliability check over images, a discipline imposed after an earlier failure in which per-image rankings from the same gradients had reliability indistinguishable from zero.

Finite-draw attenuation. Finite draws are not innocuous here. Every finite-draw cosine is attenuated below its population value, and the attenuation is arm-dependent: per-draw gradient variance grows as the token count falls, the MSE loss averaging over fewer elements. The natural estimate of Eq. 1, a redraw floor minus a single demoted-arm cosine, is also asymmetric: the floor compares *two* finite-draw estimates of the native gradient where each demote arm contributes one. An iso-direction null, in which demotion changes *nothing* but the variance, therefore still produces a positive gap that *grows as the target grid shrinks*, the same signature as "absolute token count sets the floor," and no naive read can tell the two apart. The mechanism is live and large at our operating point: a variance back-of-envelope predicts spurious iso-direction endpoint gaps of $\approx 0.02/0.05/0.15$ for 896/768/512, and the uncorrected 1024→896 endpoint gap indeed decays as $+0.100/+0.035/-0.016$ at $D = 4/8/16$, a clean c/D decay (Appendix B).

The debiased estimator. Two corrections, used together, remove the finite-draw attenuation. First, *self-floors with attenuation correction*: every arm runs a second independent draw set g'_a, giving a per-arm self-cosine $\cos_{\text{self}} = \cos(g_a, g'_a)$, so that each arm has the same two-estimate structure the native floor has. The debiased cosine is the classical split-half disattenuation correction,

$$\hat{c} \;=\; \frac{\cos(\bar{g}_{\text{src}},\, \bar{g}_a)}{\sqrt{\cos_{\text{floor}} \cdot \cos_{\text{self},a}}}, \qquad \widehat{\text{gap}} \;=\; 1 - \hat{c}, \tag{7}$$

which cancels the finite-draw attenuation of numerator and denominator to first order. Second, *draw-count extrapolation*: the attenuation of an accumulated-draw cosine decays as $1/D$, so sweeping nested draw subsets and fitting $\text{gap}(D) = \text{gap}_\infty + c/D$ recovers the draw-limit gap directly. The two are mutually checking, and the check passes: debiased fits come out D-flat ($|c| \le 0.05$ on the 512 route, against $c \approx +0.29$ uncorrected), and the nested sweep extrapolates the native redraw floor to a self-cosine of 1.005 (bootstrap 95% CI $[0.994, 1.016]$). That is, the population per-bin native gradient is a *fixed direction*, and the entire redraw floor is finite-draw noise; remaining validation detail is in Appendix B.

Margins and resolution. A safety verdict needs two numbers, stated separately: a margin, the largest excess considered practically equivalent and a property of the question, and the instrument's resolution, which sets only the power available to certify it. We fix the margin at $\varepsilon = 0.02$, comparable to the re-encode control's own confidence half-width (Table 7). A route is *certified safe at margin* ε in a bin exactly where the one-sided non-inferiority bound clears it,

$$\text{UB}_{95}(\overline{\text{gap}}) \;<\; \varepsilon, \tag{8}$$

and a *window* claim over several bins requires the simultaneous (Bonferroni-corrected) bounds to clear it in every bin. For a route whose true excess is zero the paired estimate $\overline{\text{gap}}$ fluctuates with standard error $\text{SE}(N, D, \sigma\text{-bin})$, so $\varepsilon^* = 1.645\,\text{SE}$ is the *certification resolution*: a cell with $\varepsilon^* > \varepsilon$ is *underpowered* at the margin, and a crossing there is reported but not counted as certification. "No certifiable window" means a route's simultaneous upper bounds never fall below ε at the current (N, D); "certifiably unsafe" means the lower bound stays above it. At the verdict grid's operating point of N=40 and D=12 per bin, the bin-level ε^* is 0.02–0.07, so interior verdicts below are stated at the margins they do certify; endpoint-mode draw sweeps, with D up to 64, reach $\varepsilon^* \approx$ 0.01–0.02, powered for the strict margin. All verdict maps report the *paired per-image excess* $\text{gap}_{e,i} = \widehat{\text{gap}}_{e,i} - \widehat{\text{gap}}_{\text{reenc},i}$, trimmed of non-finite and $|\text{gap}_{e,i}| > 1.5$ values, which cancels the shared per-image draw structure and stays stable at mid-σ where the unpaired estimator is not; finite-draw cosines are compiler-kernel-path sensitive, so all pairings are computed within one run. Every number in the main text is debiased. Four secondary probes (the 1280-tier iso-severity twin, the 896-tier transfer, the depth split, and the positional-interpolation intervention) predate the debiased instrument and are retained only where their strictly positive bias is conservative or cancels in a paired same-grid contrast (Appendices B, F).

4.2 The measured (route, σ) map

We measured the map directly: the measured (black) debiased gap curves of Fig. 1 carry the verdict, read per bin against the gray $\pm\varepsilon^*$ band per Eq. 8 (all three routes on one axis, with the estimator context, in Appendix Fig. 7; per-bin numbers in Appendix Table 6); §4.3 scores both accounts on the same curves. Throughout, *safe* means exactly the certification of Eq. 8, namely gap excess over the re-encode control below the stated margin, on the *per-example* object: the sense in which a demoted gradient is indistinguishable from a native one. Route by route: 1024→896 (floor 0.02–0.04) is

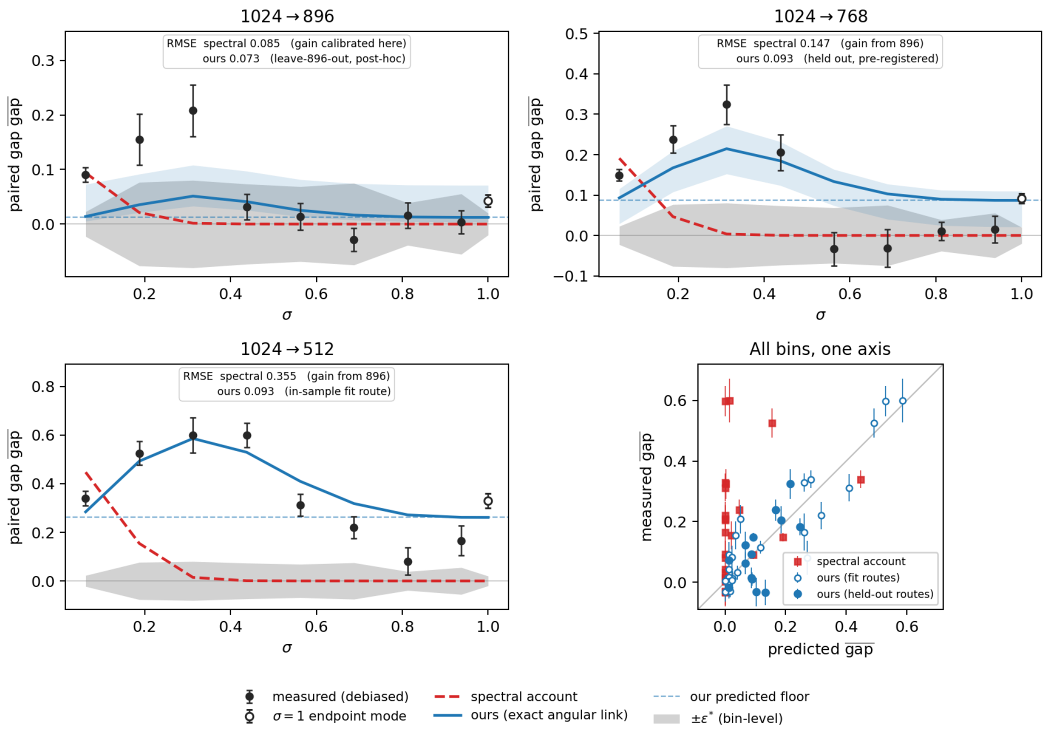

Figure 1: **The measured map, and both accounts on the same axes.** Per route, one set of measured debiased gap curves (Eq. 1, black; open markers detach the σ=1 endpoint bin, a distinct probe mode) carries the verdict of §4.2, read per bin against the gray band, the bin-level $\pm\varepsilon^*$ certification resolution (§4.1). The same axes carry both predictions of §3: the spectral account (red dashed), transported through the residual→gradient bridge at the measured anchor tolerance, the whole family scoring alike (Appendix Fig. 5); and ours under the exact angular link (blue, Eq. 18; three-form comparison in Appendix Fig. 3), with each route's predicted floor (blue dashed; dashed-to-solid is the data term) and a 95% full-pipeline bootstrap band (B=1000; Appendix B). Neither account is fit to the route it is drawn on except as labelled in-panel: ours predicts 1024→768 from governors alone with zero per-route freedom (pre-registered held-out) and 1024→896 under the post-hoc leave-896-out re-derivation, while the spectral account's single gain is calibrated on 1024→896 and frozen. The 768 mid-σ dip and the 896 low-σ plateau sit outside the 95% band; these are the two edges of the reduction's domain (Appendix C). Bottom right: predicted vs measured over every bin, with ours also on the two 1280-tier routes the spectral bridge does not cover.

safe on the interior window $\sigma \in (0.5, 1)$, certified at margin 0.09 simultaneously over the seven window bins. 1024→768 and 896→768 (floors 0.06–0.13) certify no window at the strict margin, but 1024→768 retains a narrow **low-excess window** $\sigma \in (0.65, 0.95)$, with paired excess +0.02–0.04 per bin, certified at margin 0.09 simultaneously over its four window bins (0.072 pointwise), atop a floor no noise level removes; this is the window §5 stacks as a second route. Any route to 512 (floor $\approx$0.30) is *certifiably unsafe* in all fifteen bins. Safety is a per-route boundary, jointly determined by the ratio, which acts through a_e, and by absolute target size, which acts through the floor. Debiasing itself moved verdicts in both directions, and the two revisions against the historical record are part of the result: roughly half the 768 route's high-σ plateau was estimator bias, softening "unsafe at every noise level" into the narrow window just stated, while the 896 endpoint shows a small *real* gap (+0.034 ± 0.011) that single-estimate variance had buried.

The map depends on the aggregation object. The aggregation operator is part of the estimand (§3.1, Appendix A.1), and the two objects genuinely disagree at high σ: pooled-arm probes at pool size 4 collapse the 1024→896 and 1024→768 excesses to $\approx$0 over the upper bins, so the batch-aggregate object SGD follows is more forgiving here than the per-example read we publish as the verdict. The coherent drift no batch size removes is bounded only by an $a + b/B$ fit across pool sizes, the designated readout of a debiased verdict grid at the trainer's actual batch×accumulation operating point **[pending]**.

4.3 Both accounts, scored

Both accounts are scored against one measured object: the debiased gap curves of Fig. 1, read as §4.2 has just set out. The spectral account is scored first, at the boundary and then at the curve level; ours follows, term by term.

The spectral account: boundaries and curves. The Qwen-Image VAE's latents are high-frequency-quiet ($P(f) < 1$ above $f \approx 0.16$ cycles/latent-pixel; latent variance 0.42), giving equal-power (input-SNR) crossovers of $0.136/0.146/{\approx}0.20$ for 896/768/512: at that tolerance demotion would be safe over almost the entire σ axis. Table 5 (Appendix D) sweeps the whole tolerance family, and no row reproduces the measured map (§4.2). At the curve level (Fig. 5, the family transported to full curves) the failure is sharper: the predicted mismatch is concentrated below $\sigma \approx 0.3$ and the transported curve is near zero exactly where the measured curves peak (RMSE 0.147 at 768 and 0.355 at 512), while above each tolerance's committed boundary, where every member predicts exactly zero, the floor persists (committed-region RMSE 0.027/0.049/0.218 for 896/768/512). Every tolerance scores within 0.01 RMSE of every other, so no conclusion hinges on the choice of δ (Appendix D). Both reads are insensitive to the residual-shape choice in the transport.

Ours, the data branch: a route-uniform numerator over a U-shaped denominator. Measuring the mean prediction residual $\bar{r} = \mathbb{E}_\epsilon[\hat{v} - (\epsilon - x)]$ per grid and σ (the exact r in $g = J^\top r$; by Eq. 14 intrinsic posterior bias plus the frozen model's approximation error, not model error alone) shows a strong, monotone σ-shape (cross-grid excess $0.89 \to 0.36$ in relative L_2 from σ=0.125 to 1) that is the *same for every route* within ±0.02, across routes whose gradient floors span 0 to 0.3 (Appendix Table 12). This result refutes the residual-level prediction of §3.2: with no cross-frequency coupling, that mismatch is confined to the destroyed band and ordered by its energy, which differs substantially across these routes. Route-uniformity implies that the real mismatch is carried by cross-band structure the diagonal model cannot express, and that all route identity lives in J. This was a pre-registered contrast that would equally have falsified our own assumption (iv); instead it *establishes* the shape factorization directly. The decay is smooth, Wiener-like shrinkage with no hard gate at the spectral crossover. The denominator then does real work: the total gradient norm is U-shaped in σ ($2.6 \to 0.4$ at $\sigma \approx 0.3 \to 9.3$; Appendix Fig. 7b), and a monotone absolute mismatch over a U-shaped norm yields exactly the measured interior-peaked curves: bin-mean gap tracks $1/\|\bar{g}_{\mathrm{src}}\|$ (Spearman +0.55–+0.90; regime-by-regime reading and the renormalization check in Appendix B). This shape read is a post-hoc consistency analysis and is labeled as such; all safety criteria stay in cosine units, since direction is what a normalized optimizer consumes.

The graph branch: the floor exists (three probes, one number). At the $\sigma = 1$ *endpoint* the input is pure ϵ, identical in distribution across arms, so the input-mediated data term is zero by construction; the debiased endpoint floors are +0.019/+0.056/+0.304 for 896/768/512 (Appendix Table 7), and the 512 value clears the pre-registered confirmation bar ($\widehat{\mathrm{gap}}_\infty \geq 0.15$) on 12 of 12 probe images individually. Per §3.3 the target-mediated share need not vanish there, so two further probes close the step. The *x-zero* probe (§4.1) removes the image from input *and* target, isolating pure graph-shape sensitivity, and its debiased draw-limit gaps equal the endpoint gaps at every route. The *target-strength sweep* re-runs the endpoint with target $\epsilon - \alpha x$, $\alpha \in \{0, \ldots, 1\}$, and the paired debiased excess is α-flat at the well-conditioned anchors (e.g. 768: $+0.070 \pm 0.015$ at α=0 vs $+0.049 \pm 0.010$ at α=1; control slope +0.003): no resolvable target-content share *in the angular estimand*. The flatness is not a cancellation mystery. The target-mediated gradient itself is real and large, but demotion's change to it lands almost entirely *along* $\hat{g}_{\mathrm{src}}$, to which the cosine gap is blind (Eq. 18); the endpoint gap is therefore the graph floor as the angular estimand reads it (parallel-landing decomposition, with the $\kappa_\parallel/\kappa_\perp$ numbers, in Appendix B). Finally, a forward-only probe of the model's caption-conditioned *prior* across grids gives route distances that are flat where the floors are strongly ordered, placing the ordering in J rather than in a resolution-conditioned prior or a training-distribution discontinuity.

Table 1: Weight-space footprint vs. the seed lottery: $\cos(\Delta W_{\text{arm}}, \Delta W_{\text{native}})$ at matched training seed, mean ± SD over three seed pairs, on two single-artist corpora (60/15 training images). The cross-seed row is the same cosine between native endpoints differing *only* in training seed; an arm above it displaces the endpoint *less* than a seed change. Training is bit-exact deterministic (an identical retrain reads 1.000); the non-det. row retrains native at the same seed *without* deterministic kernels (one twin pair per corpus), so it reads run-to-run chaos alone. *late*: demotion only in the last 75% of steps; the stacked row adds a second certified route (1024→768 on $0.65<\sigma<0.95$, last 50%). **Bold**: closest endpoints.

arm (vs. native, matched seed)	corpus A	corpus B
native, cross-seed	0.121 ± .007	0.199 ± .003
native, same-seed non-det. retrain	0.264	0.427
896, σ>0.5, aligned	0.365 ± .020	0.432 ± .026
+ late	**0.753 ± .019**	**0.770 ± .009**
+ late, stacked 768	**0.753 ± .017**	**0.771 ± .009**
896, every σ	0.183 ± .009	0.236 ± .009

Table 2: The proposed step, in full. Lines 3–5 (colored) are the entire change: the gate σ^*=0.5 and the route are read off the measured map (§4.2), line 5 is the optional refinement, and lines 1–2 are only a reordering of a standard step, such that σ is drawn before the latent is fetched.

one LoRA training step

1 $\sigma \sim p_{\text{train}}$
2 $x \leftarrow$ `native latent of` y
3 `if` $\sigma > \sigma^*$ `and` y `on-route:`
4   $x \leftarrow$ `demoted latent of` y
5   `rotary` $\leftarrow$ `banded(`σ`)`
6 $\epsilon \sim \mathcal{N}(0, I)$ `at shape(`x`)`
7 $z \leftarrow (1-\sigma)x + \sigma\epsilon$
8 $v \leftarrow \epsilon - x$
9 `step on` $\|\hat{v}_\theta(z, \sigma, c) - v\|^2$

The governors: ratio sets the amplitude, absolute size sets the floor. With the floor isolated, what orders the routes? Two probes give a clean division of labor, and together they decide the equal-ratio disagreement, limit (b) of §3.2. *Absolute size, not ratio, sets the floor*: re-run one tier down, the route 896→768 fails its pre-registered bar with a high-σ residual ~2× the 1024→896 plateau despite a near-identical downscale ratio, while the two routes sharing the ~2,160-token target grid, 1024→768 and 896→768, land on the same floor despite different ratios (Appendix Table 11). Near-equal ratio with a different target gives a different verdict; the same target with a different ratio gives the same floor. This refutes any pure ratio rule for the floor, and with it the spectral account's only degree of freedom. Moreover, the floor grows monotonically as the *target* grid shrinks (0.02–0.04 / 0.06–0.09 / ≈ 0.30 at target ~3,000/2,160/1,000 tokens), consistent with the reading that the floor measures how well the coarse graph approximates the fine graph's computation. *Ratio, not absolute size, sets the amplitude*: a synthetic iso-severity route 1280→1120, with edge ratio exactly matched to 1024→896 at 1.6× its absolute target size, reproduces the 1024→896 curve bin-for-bin, crossover included (Appendix Table 10). Note what survives of the spectral account here: ratio *is* the right governor for the data term; its error is not its governor but its scope, mistaking one term for the whole gap.

The floor's own parts: $\text{RoPE}_e + \text{Resid}_e$. The prediction of §3.3, that the positional share is a coordinate-system artifact, is testable by an origin-side intervention: evaluate the demoted grid at positionally-interpolated fractional rotary coordinates (patch i at $i \cdot H_{\text{src}}/H_{\text{dem}}$), making its relative phase geometry exactly match native. At the σ=1 endpoint this *erases* the 768 floor and removes ~30% of the 512 floor, leaving the in-band 896 control untouched: re-anchored against the debiased floors, $\text{Resid}_{768} \approx 0$ while 512 retains a $\text{Resid}_{512} \approx 0.2$ bulk. The mild-route floor is mostly a coordinate-system artifact; the harsh-route bulk is a genuine non-positional graph residue, and the capacity governor attaches to it, as the band-counting sketch anticipated. The erasure is *not* a training lever, since with image content in the input the stretched forward is off-manifold, but a frequency-*selective* banded alignment survives in-window (§5). Full numbers, the per-block/per-module depth localization behind this design, and the inference-side prior art on the attention-over-N piece (Li et al., 2025; Liu et al., 2026) are in Appendix B.

5 σ-CONDITIONAL TRAINING ON THE MEASURED-SAFE ROUTE

We wire the one measured-safe corpus route into the LoRA trainer as an opt-in path, exactly as measured, with no extrapolation beyond the map. The intervention is deliberately small: three lines inside an otherwise standard flow-matching step, and no change to the objective, the optimizer, or the noise density (pseudocode in Table 2).

σ-first draw. The trainer draws each batch's σ *before* fetching latents, from the unchanged training density (demotion conditions on σ, not the reverse). When every sample in the batch is above the gate ($\sigma > 0.5$), the native 1024-tier latent is swapped for a cached demoted-grid sibling produced by the probe's own measured-safe recipe (pixel-space downscale, VAE re-encode); everything downstream derives from the swapped latent. Off-route images always train native, as does validation. One asymmetry between the gate and the map must be stated: the gate is one-sided ($\sigma > 0.5$) while certification stops at $\sigma = 0.94$, and the endpoint carries a small real gap ($+0.042 \pm 0.011$). Since the expected gradient should vary continuously in σ, that gap cannot be dismissed as a measure-zero boundary point: it implies a neighborhood, of unmeasured width, over which some of it survives, and the tail $\sigma \in (0.94, 1)$ is exactly the interval the verdict grid does not resolve. Two facts bound the exposure meanwhile: the last interior bin reads $+0.004 \pm 0.021$, and the exercise grid of Table 1 trained *with* the tail demoted and still landed a factor of 2–3 inside the seed lottery, so any tail effect is below that instrument's resolution at 480 steps. A dense upper-tail sweep, covering $\sigma \in (0.9, 1)$ in four bins plus the endpoint at higher draw count, is designated to resolve the shape. Until it lands, the certified condition is $\sigma \in (0.5, 0.94]$, and the unbounded gate should be read as the form we exercised, not the form the map certifies.

Banded rotary alignment, σ-gated. Motivated by §4.3, a frequency-selective alignment is applied on demoted forwards only. It uses YaRN-style bands: a full positional-interpolation stretch for rotary bands with few rotations across the demoted extent, native spacing for high-frequency bands, a ramp between, and thresholds scheduled by a sigmoid gate in σ following Zhao et al. (2026). Unlike uniform interpolation it is *not* off-manifold in-window: it passed both pre-registered legs, erasing the static variant's low-σ liability ($+0.064 \rightarrow +0.033 \pm 0.025$) while preserving its high-σ gains (paired -0.05 vs plain demotion at $\sigma \geq 0.59$). It is a refinement candidate, not a gate-widener: the gate stays $\sigma > 0.5$. Under bit-exact deterministic training it adds no weight-space displacement over the demotion arm it refines (base↔refined 0.319 vs base↔demoted 0.320; without determinism a twin-run chaos floor of 0.41 would swamp this read).

Cost accounting. With ~50% of draws above the gate and the demoted forward at ~0.72× tokens, the fixed-step epoch cost is ≈0.86, a *projected ceiling* of ~14% wall-clock on our corpus, where 96% of records are on the safe route; the ceiling scales with high-tier content. On the fixed-step exercise grid below, the realized saving is −15.1% forward FLOPs (exact operator-level counts over each run's realized token histogram) and −14.6% wall-clock (n=3 seed means on two single-artist corpora; wall-clock tracks FLOPs ≈ 1:1): the projection is achieved. The claim is wall-clock at fixed steps, not "more steps in the same time."

Weight-space endpoint footprint. A fixed-step exercise grid (gated, scheduled, and gate-removed arms × 3 training seeds × 2 single-artist corpora, 480 optimizer steps each, bit-exact deterministic paired-RNG so every arm sees identical draws) compares the trained adapters directly in weight space. Table 1 reads the cosine between each arm's endpoint ΔW and its matched-seed native twin's, judged against a seed-noise yardstick: the same cosine between native endpoints that differ only in training seed. Determinism makes the read exact: an identical retrain reproduces its twin at 1.000, whereas the same retrain with non-deterministic kernels reads 0.264/0.427 (corpus A/B; run-to-run chaos alone; second row of Table 1). The yardstick is tight and arm-independent, with the gated arm's cross-seed values coinciding with it within .006, so the margins are many seed-spreads wide. Every arm lands above the yardstick: even demoting *all* 480 steps displaces the endpoint less than changing the training seed, and the σ>0.5 gate sits a factor of 2–3 above it (0.365 vs 0.121; 0.432 vs 0.199); on corpus B it coincides with the non-deterministic retrain (0.427), i.e. it sits within kernel-noise reach of a native retrain. The ordering, however, is set by *placement*, not by per-step severity. Protecting the first quarter of training doubles the closeness (0.365 → 0.753), consistent with early subspace selection in a from-zero adapter, and stacking the strictly deeper 768 low-excess route onto the late rule is endpoint-free (0.753, 0.771), although the spectral account of §4.3 prices every 768 substitution as more severe than 896. A per-step account is placement-blind by construction; the endpoint statistic resolves what it cannot: *when* a demoted step happens dominates *how deep* it goes, provided the route stays inside the certified window. Removing the σ-gate erodes the margin toward the lottery radius (0.183; 0.236). Weight-space proximity is a footprint statement, not a quality statement: sample-level orderings need not track it, and generation-quality non-inferiority at production scale remains the stated open gate (§6).

Figure 2: Visual counterpart to Table 1: matched (prompt, noise-seed) renders from each arm's endpoint on a corpus-B display prompt, 20 steps; columns are the table's arms, rows the three training seeds. Moving along a row (arms, seed fixed) changes the render on the same visual order as moving down a column (seed lottery, arm fixed); the gate-removed arm (rightmost) drifts the most, matching its eroded margin in Table 1.

6 Limitations

One model family, one operating point. All gradient probes ran on one DiT (Anima) with one adapter checkpoint trained at native tiers. A mixed-resolution-trained adapter might equalize (or widen) its own gradients; probing such a checkpoint is the designated bound on how far the map generalizes, as is re-probing 1280→1024 after fine-tuning at the 1280 tier. The probe pool's relationship to the adapter's own fine-tune set was measured rather than conceded: a controlled 2×2 factorial (two LoRAs trained on opposite style clusters under frozen membership manifests, probed on both) replicates the map's shape on both checkpoints with the adapter×probe-style interaction below instrument resolution, while the absolute redraw-floor level is checkpoint-dependent (Appendix E).

Gradient-level, not end-task. The map is built from accumulated per-step gradients; integration over an optimizer trajectory is addressed by the exercise grid's weight-space read at 480 steps rather than production scale, and the generation-quality (CMMD, Jayasumana et al., 2024) gate remains open: the exercise grid's first scoring pass could not separate even the off-map control from native at this scale, so we read no quality verdict from it.

Account resolution and domain. The account predicts held-out routes at ~0.07–0.09 RMSE, not within ε^*; its $A(\text{ratio})$ governor rests on two distinct ratio values; the reduction's domain excludes the 768 mid-σ window, whose mechanism the vector-resolved probe has measured (negative interference) but which the two-term form still does not predict; the floor law's functional form is unidentified at our operating points; and the band-counting sketch for $\mathrm{RoPE}_e/\mathrm{Resid}_e$ is calibrated, not derived. The graph-floor reading of the endpoint is also estimand-specific: the target-mediated perturbation is real at the vector level and lands parallel to the native gradient (Appendix B), so a magnitude-sensitive estimand, such as an un-normalized optimizer, could apportion the endpoint gap differently than the angular read does.

Instrument resolution and coverage. "Safe" is one-sided non-inferiority against a fixed margin (ε = 0.02 strict; the interior window currently certifies at 0.056–0.07), with the certification resolution (ε^* = 0.03–0.08 per bin; ≈ 0.01–0.02 at the endpoint sweeps) setting the power available, not the margin. True gaps below ε^* are undetectable by construction, σ^* boundaries are localized only to a bin, and every "no certifiable window" claim is (N, D)-relative: a larger campaign could certify (or refute) the 768 high-σ window at the strict margin. The upper tail $\sigma \in (0.94, 1)$ of the safe route is unmeasured between the last interior bin and the endpoint's real gap; the designated dense sweep (§5) closes it. Verdict stability under the per-image trim rule of §4.1 has not been separately audited. The debiasing campaign covered the corpus verdict grid; the 1280-tier iso-severity, 896-tier transfer, depth-split, and PI-intervention probes remain pre-debiasing reads, and the 1280-tier curves enter the held-out validation on that basis.

Scope. Adapter (LoRA) gradients on a frozen backbone; full fine-tuning may differ. Pixel-space demotion only (latent-space downsampling was not tested here; SwD found it inferior). The σ-gated banded-alignment parameters were taken from one shot at pre-registered values, not searched.

7 Conclusion

When does training on downscaled images yield the same gradient direction? Not when the noise has masked the frequencies the coarse grid loses. That spectral answer was advanced by prior work for inference-time substitution and had not previously been confronted with the training gradient. Transported to the training question, it is a model of only one part of one branch of what demotion perturbs, and the measurements reject it at every point of structural divergence: the mildest route's crossover sits at 3.5× the equal-power prediction, no tolerance reproduces the map, and a 2×-downscaled latent certifies at no noise level, because the network function itself is grid-calibrated, partly through its positional coordinate system and partly through an irreducible dependence of the computation on token count.

The measured answer has two conditions, matching the two terms of our account: the target grid must be large enough in *absolute token count* that its graph floor sits below certification resolution, and the noise level must exceed a per-route boundary where the ratio-governed data term dies. On our model that means 1024→896 on $\sigma \in (0.5, 0.94]$ (the upper tail toward the endpoint's small real gap still unresolved) and 1280→1024 at $\sigma \gtrsim 0.75$, and a 2× downscale at no noise level. Between the regimes, 1024→768 keeps a narrow low-excess window ($0.65 < \sigma < 0.95$) atop its floor, low enough to stack onto the trainer at no endpoint cost, short of strict certification. The answer is predictive, not merely descriptive: head to head on the same curves the account predicts held-out routes at ~0.07–0.09 RMSE against the spectral family's 0.147–0.355, beating an oracle fit on the held-out data itself on one route. Getting there required treating the question as a metrology problem: naive finite-draw estimation manufactures exactly the token-count-ordered floor signature under test, half the intermediate route's raw floor was estimator bias, and our own headline claims were re-derived, and in one case softened, under the correction. On the one route our corpus makes safe, the trainer realizes the map's projected ceiling of −14.6% wall-clock at fixed steps, with a weight-space endpoint footprint a factor of 2–3 inside the seed lottery, rising to cosine 0.75 against the native endpoint when demotion is late-scheduled; quality non-inferiority at production scale remains the open gate. We offer the account and its instrument as the cheap, general test that any future "train part of the time at lower resolution" proposal should pass before it is believed. Without the debiasing, that test manufactures the very effect it looks for.

Reproducibility statement

The probe scripts (`run_sigma_probe.py`, including the `--self_floor` and `--draw_sweep` debiasing modes; `run_prior_distance.py`, `rapsd.py`, `compare_ckpt_dw.py`), the held-out validation and refit analyses (`e5_holdout.py`, `e5_refit.py`, `e83_bridge.py`), the trainer wiring (`--sigma_lowres`), and the frozen pre-registration documents are in the public repository at `https://github.com/sorryhyun/anima_lora` (project line `project/sigma_lowres/`), together with the debiased verdict runs' complete per-image records. The earlier raw-instrument archives are excluded by the repository's results ignore rule and will be published as a companion tarball **[pending]**; until then those numbers are reproducible from the scripts but their original records are not public. Paired training runs use a `--deterministic` mode verified bit-identical across twin runs; finite-draw cosines are kernel-path-sensitive, so all cosine pairings are computed within a single run (Appendix B).

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A Notation and derivations

A.1 Notation and estimand fine print

Arm labels versus route subscripts. One notational convention holds throughout. Arm labels (src, reenc, dem) mark *whose* gradient, latent, or Jacobian an object is; the subscript on every route-level scalar is the *route*, written in full ($d_{e_0\to e}$) where several sources appear, and abbreviated to the target edge alone wherever the source edge is fixed by context, as it is in almost every measured statement; the abbreviation is an edge value, never an arm label, so it always instantiates numerically (d_e, Φ_{1120}, Resid_{768}). The control's distance d_{reenc} (Eq. 1 with $\bar{g}_{\mathrm{reenc}}$ in place of $\bar{g}_{\mathrm{dem}}$) is the one exception that carries an arm label: the control changes no grid and has no target edge. Each table states which object it reports, d_e or the excess gap_e.

The two aggregations. In a batched training scenario Eq. 1 admits two aggregations over a probe set: per-example (cosine per image, then mean; the object every map in the body reports) or batch-aggregate (gradients summed over the batch before the cosine; the object SGD actually follows). These are different estimands, since the aggregation operator is part of the estimand, and no coefficient fitted on one is guaranteed to transfer to the other. At second order the two are related by a batch-size decomposition: writing each image's demotion-induced gradient disagreement as a coherent mean b plus a zero-mean idiosyncratic deviation with covariance Σ_η, the batch-aggregate distance at batch size B is

$$\mathbb{E}[d_B] \;\approx\; \frac{\|P^\perp b\|^2}{2\|\mu\|^2} \;+\; \frac{\mathrm{tr}\big(P^\perp \Sigma_\eta P^\perp\big)}{2B\|\mu\|^2},$$

with μ the mean source gradient and $P^\perp$ the projector off $\hat{\mu}$: an intercept (the coherent share, which never averages out) plus a $1/B$ term. Monotone improvement with B is not automatic, since it would require an iid zero-mean disagreement model we have not verified, but where we have measured the batch object it is the more forgiving of the two: pooled-arm probes at pool size 4 collapse the 1024⟶896 excess to ≈ 0 at every bin $\sigma \geq 0.625$ and the 1024⟶768 excess at $\sigma \geq 0.875$ (§4.2), the $1/B$ term dominating there. What the collapse does not bound is the intercept, exactly the coherent drift that the intercept-plus-$1/B$ fit across pool sizes (§4.2) is designated to bound.

A.2 Derivation of the branch decomposition (Eq. 5)

Fix a query (image, caption), a σ, and an arm a. Per draw ϵ, the adapter gradient of the $\frac{1}{2}$-scaled loss is the product $g_a(\epsilon) = J_a(\epsilon)^\top r_a(\epsilon)$ of §3.3. Split each factor into its noise mean and fluctuation, $J_a = \bar{J}_a + \tilde{J}_a$ and $r_a = \bar{r}_a + \tilde{r}_a$ with $\mathbb{E}_\epsilon[\tilde{J}_a] = 0$ and $\mathbb{E}_\epsilon[\tilde{r}_a] = 0$; the population mean gradient of arm a is then

$$\bar{g}_a \;=\; \mathbb{E}_\epsilon\big[J_a^\top r_a\big] \;=\; \bar{J}_a^\top \bar{r}_a \;+\; \mathbb{E}_\epsilon\big[\tilde{J}_a^\top \tilde{r}_a\big]. \tag{9}$$

Write the demoted arm's mean factors as native plus a perturbation, $\bar{J}_{\text{dem}} = \bar{J} + \Delta J$ and $\bar{r}_{\text{dem}} = \bar{r} + \Delta\bar{r}$ (bars without an arm subscript denote the native arm). Subtracting Eq. 9 across arms and expanding the product,

$$\delta g \;=\; \bar{g}_{\text{dem}} - \bar{g}_{\text{src}} \;=\; \underbrace{\bar{J}^{\top}\Delta\bar{r}}_{\text{data branch } B_e} \;+\; \underbrace{\Delta J^{\top}\,\bar{r}}_{\text{graph branch } C_e} \;+\; \underbrace{\Delta J^{\top}\Delta\bar{r} \;+\; \mathbb{E}_{\epsilon}\big[\tilde{J}_{\text{dem}}^{\top}\tilde{r}_{\text{dem}}\big] - \mathbb{E}_{\epsilon}\big[\tilde{J}_{\text{src}}^{\top}\tilde{r}_{\text{src}}\big]}_{\text{remainder } R_e}. \quad (10)$$

This is an exact identity: no linearization is involved. "First-order" in the main text refers only to the content of R_e: it collects every term that is a product of two perturbations ($\Delta J^{\top}\Delta\bar{r}$) or a difference of noise-covariance terms (the last two). Eq. 5 is Eq. 10 with the mean-Jacobian bar suppressed in the notation.

One remark delimits the identity's literal scope. When the two arms share a grid (as for the re-encoding control), every operation above is literal. Across grids, J_{src} and J_{dem} act on different token counts, so the differences ΔJ and $\Delta\bar{r}$ require a fixed identification of the two output spaces (e.g. spectral zero-padding of the coarse grid onto the fine grid's bands); any such identification changes the split between B_e, C_e, and R_e but not their sum δg, which lives in the shared adapter-parameter space regardless. This is why the main text defines the branches *operationally* (as the two independent perturbation directions in adapter-parameter space, isolated by the designated probes of §4.3) and treats Eq. 5 as fixing what each branch collects rather than as a computable matrix formula.

A.3 The mean residual as a posterior operator

This appendix derives the posterior form of the fixed-image mean residual, the exact split of the measured object into intrinsic posterior bias plus model approximation error, and the Gaussian closure referenced in §3.3. Fix a caption c and grid e, write $a = 1 - \sigma$, and let $x \sim p_e(\cdot \mid c)$ be the grid-e clean latent, $z_\sigma = ax + \sigma\epsilon$ with $\epsilon \sim \mathcal{N}(0, I)$.

Bayes field and fixed-image residual. For the squared objective the population-optimal field is the conditional mean of the target $v = \epsilon - x = (z_\sigma - x)/\sigma$:

$$v_e^*(z, \sigma, c) \;=\; \mathbb{E}[v \mid z_\sigma = z,\, c] \;=\; \frac{z - m_e(z, c)}{\sigma}, \qquad m_e(z, c) := \mathbb{E}[x \mid z_\sigma = z,\, c]. \quad (11)$$

Averaging over noise at fixed image ($\mathbb{E}_\epsilon[\epsilon] = 0$, $\mathbb{E}_\epsilon[z_\sigma] = ax$) gives the posterior form used in §3.3:

$$\bar{r}_e^*(\sigma; x) \;=\; \mathbb{E}_\epsilon\big[v_e^*(ax + \sigma\epsilon, \sigma, c) - (\epsilon - x)\big] \;=\; \frac{x - \mathbb{E}_{z_\sigma\mid x}\big[m_e(z_\sigma, c)\big]}{\sigma}. \quad (12)$$

With $p_{\sigma,e}(z \mid c)$ the noised marginal, Tweedie's identity for the linear interpolant reads $m_e(z, c) = \big(z + \sigma^2\nabla_z \log p_{\sigma,e}(z \mid c)\big)/a$ for $\sigma < 1$, so equivalently

$$\bar{r}_e^*(\sigma; x) \;=\; -\frac{\sigma}{1-\sigma}\,\mathbb{E}_{z_\sigma\mid x}\big[\nabla_z \log p_{\sigma,e}(z_\sigma \mid c)\big], \quad (13)$$

which makes the missing ingredient explicit: the numerical curve requires the grid- and caption-conditional smoothed score, i.e. a model of $p_e(x \mid c)$. At $\sigma = 1$, where $z = \epsilon$ is independent of x, $m_e(z, c) \to \mathbb{E}[x \mid c]$ and $\bar{r}_e^*(1; x) = x - \mathbb{E}[x \mid c]$. With A_e the alignment operator that places the source-grid residual on the demoted grid (§4.1), the Bayes-level cross-grid mismatch is $\Delta\bar{r}_e^*(\sigma; x) = \bar{r}_{\text{dem}}^*(\sigma; x_{\text{dem}}) - A_e\bar{r}_{\text{src}}^*(\sigma; x_{\text{src}})$.

What the instrument measures. MSE optimality gives $\mathbb{E}[v^* - v \mid z, c] = 0$ and hence a zero residual after a *joint* average over (x, ϵ); it does not zero the fixed-image average $\mathbb{E}_\epsilon[v^* - v \mid x, c]$, which is Eq. 12 and generally nonzero. Writing the trained network as $\hat{v}_{\theta,e} = v_e^* + q_{\theta,e}$, the probe's per-image object splits exactly as

$$\bar{r}_{\theta,e}(\sigma; x) \;=\; \bar{r}_e^*(\sigma; x) \;+\; \mathbb{E}_{z_\sigma\mid x}\big[q_{\theta,e}(z_\sigma, \sigma, c)\big] \;: \quad (14)$$

intrinsic posterior reconstruction bias, derivable in principle from $p_e(x \mid c)$, plus genuinely model-specific approximation error. The measured $\|\Delta\bar{r}(\sigma)\|$ therefore has definite theoretical status, as an empirical closure of Eq. 12, but is not a pure model-error difference, and no claim in the paper requires it to be one: the two-term account consumes the measured curve as-is (assumption (iv)), and the trained-model residual is in any case the exact r-factor that the LoRA gradient $g = J^{\top}r$ is built from.

Gaussian closure. The cheapest nontrivial closure is a full Gaussian model $x \mid c \sim \mathcal{N}(\mu_e, C_e)$ per grid. With $Q_e(\sigma) = a^2 C_e + \sigma^2 I$, the posterior mean is linear, $m_e(z, c) = \mu_e + aC_e Q_e^{-1}(z - a\mu_e)$, and substituting into Eq. 12 collapses (via $I - a^2 C_e Q_e^{-1} = \sigma^2 Q_e^{-1}$) to

$$\bar{r}_e^*(\sigma; x) \;=\; \sigma\, Q_e(\sigma)^{-1}(x - \mu_e). \tag{15}$$

For paired source and demoted latents with covariance blocks C_{ss}, C_{dd}, C_{ds} and $L_s = \sigma Q_s^{-1}$, $L_d = \sigma Q_d^{-1}$, the image-population mean squared mismatch is

$$\mathbb{E}\big\|\Delta\bar{r}_e^*\big\|^2 = \operatorname{tr}\big(L_d C_{dd} L_d^\top\big) + \operatorname{tr}\big(A_e L_s C_{ss} L_s^\top A_e^\top\big) - 2\operatorname{tr}\big(L_d C_{ds} L_s^\top A_e^\top\big), \tag{16}$$

a σ-curve computable from paired *clean latents only*, with the source–demote cross-covariance C_{ds} load-bearing. A *diagonal Fourier C_e* makes Eq. 15 the per-band Wiener shrinkage of §3.2, confining the mismatch to the destroyed band and ordering it by destroyed-band energy: the residual-level prediction that route-uniformity falsifies (§4.3). We tested whether any structured second-order closure suffices, entirely on stored paired clean latents (no denoiser forward): three nested covariance models (diagonal spectral; + within-band channel blocks; + octave-pair cross-band coupling), fitted on held-out-from-probe images and scored on the probe set against the measured curves. All three reproduce the curve's monotone σ-shape (Pearson 0.94–0.97), its route-uniformity, the near-zero re-encoding control, and the σ=1 endpoint, but over-predict the low-σ amplitude by ∼40%, failing a pre-registered RMSE bar. Second-order data-only structure therefore recovers the shape but not the level, and the essential missing ingredients are the ones the identity names: caption-conditioning (the closure is caption-marginal), non-Gaussian posterior structure, and the q_θ term of Eq. 14. The paper therefore retains the measured $\|\Delta\bar{r}(\sigma)\|$ as its minimal honest closure. A theory–instrument comparison must also reproduce the instrument's estimand, the split-half-corrected *relative* L_2 distance of §4.1, i.e. $\sqrt{\mathbb{E}\|\bar{r}_d - A_e\bar{r}_s\|^2}$ normalized by the mean of the two arms' root-mean-square norms, rather than a raw mismatch norm.

A.4 The angular reduction: exact form and four-term expansion

This appendix states and derives the two angular forms summarized in §3.3. Both follow from Eq. 1 and the split of δg by the native direction; no property of demotion enters. Write $g = \bar{g}_{\mathrm{src}}$, $G = \|g\|$, $\hat{g} = g/G$, and $u^\perp = u - (\hat{g}^\top u)\,\hat{g}$; decompose $\delta g = G\,\kappa_\parallel\,\hat{g} + \delta g^\perp$ with $\kappa_\parallel = \hat{g}^\top\delta g/G$ and $\kappa_\perp = \|\delta g^\perp\|/G$.

The exact form. The two components are orthogonal, so $\|g + \delta g\|^2 = G^2(1+\kappa_\parallel)^2 + G^2\kappa_\perp^2$ and

$$\cos\big(g,\ g+\delta g\big) \;=\; \frac{\hat{g}^\top(g+\delta g)}{\|g+\delta g\|} \;=\; \frac{1+\kappa_\parallel}{\sqrt{(1+\kappa_\parallel)^2+\kappa_\perp^2}} \;=\; \frac{1}{\sqrt{1+\kappa_{\mathrm{eff}}^2}}\,, \qquad \kappa_{\mathrm{eff}} = \frac{\kappa_\perp}{1+\kappa_\parallel}\,, \tag{17}$$

valid whenever $1+\kappa_\parallel > 0$ (the perturbation does not reverse the gradient direction). The gap of Eq. 1 is therefore, *exactly*,

$$d_e \;=\; 1 - \frac{1}{\sqrt{1+\kappa_{\mathrm{eff}}^2}}\,, \tag{18}$$

the exact angular link of §3.3. Since $1 - \cos(\hat{g}_1, \hat{g}_2) = \frac{1}{2}\|\hat{g}_1 - \hat{g}_2\|^2$ for unit vectors, starting from either expression in Eq. 1 lands on the same result.

The quadratic limit. For small perturbations, $1 - (1+\kappa_{\mathrm{eff}}^2)^{-1/2} = \frac{1}{2}\kappa_{\mathrm{eff}}^2 + O(\kappa_{\mathrm{eff}}^4)$ and $\kappa_{\mathrm{eff}}^2 = \kappa_\perp^2\big(1 - 2\kappa_\parallel + O(\kappa_\parallel^2)\big)$, so

$$d_e \;=\; \frac{\|\delta g^\perp\|^2}{2G^2} \;+\; \underbrace{O(\kappa_\perp^2\kappa_\parallel) + O(\kappa_\perp^4)}_{\text{beyond-quadratic}}. \tag{19}$$

The four-term expansion. Substituting $\delta g = B_e + C_e + R_e$ (Eq. 10) into the leading term and expanding the square,

$$\|\delta g^\perp\|^2 = \|B_e^\perp\|^2 + \|C_e^\perp\|^2 + 2\,\langle B_e^\perp,\, C_e^\perp\rangle + \Big(2\,\langle (B_e{+}C_e)^\perp,\, R_e^\perp\rangle + \|R_e^\perp\|^2\Big),$$

whose first three terms, divided by $2G^2$, define the data share, the graph share, and the projected interaction:

$$d_e(\sigma) \;\approx\; \underbrace{\frac{\|B_e^\perp\|^2}{2\,\|\bar g_{\mathrm{src}}\|^2}}_{\text{data share } S_e} \;+\; \underbrace{\frac{\|C_e^\perp\|^2}{2\,\|\bar g_{\mathrm{src}}\|^2}}_{\text{graph share } \Phi_e} \;+\; \underbrace{\frac{\langle B_e^\perp,\, C_e^\perp\rangle}{\|\bar g_{\mathrm{src}}\|^2}}_{\text{projected interaction } I_e} \;+\; R'_e, \tag{20}$$

the four-term expansion of §3.3. The remainder R'_e has an exact content: the R_e-bearing terms above, the beyond-quadratic terms of Eq. 19, and the parallel-component correction, each suppressed by an extra power of perturbation size relative to the retained terms.

From the four-term form to the excess. The reported object is the excess over the control. The reenc arm changes no grid, so $\Delta J_{\mathrm{reenc}} = 0$, and both control terms built from it vanish identically: its graph share Φ_{reenc} and its projected interaction I_{reenc} each carry a factor $(\Delta J_{\mathrm{reenc}}^\top \bar r)^\perp = 0$. Writing Eq. 20 for each arm and subtracting term by term,

$$\begin{aligned}
\mathrm{gap}_e(\sigma) &= d_e(\sigma) - d_{\mathrm{reenc}}(\sigma) \\
&\approx \big(S_e - S_{\mathrm{reenc}}\big) + \big(\Phi_e - \Phi_{\mathrm{reenc}}\big) + \big(I_e - I_{\mathrm{reenc}}\big) + \big(R'_e - R'_{\mathrm{reenc}}\big) && \text{Eq. 20, each arm} \\
&= \big(S_e - S_{\mathrm{reenc}}\big) + \Phi_e + I_e + \big(R'_e - R'_{\mathrm{reenc}}\big), && \Phi_{\mathrm{reenc}} = I_{\mathrm{reenc}} = 0
\end{aligned}$$

so the control collapses to $d_{\mathrm{reenc}} \approx S_{\mathrm{reenc}} + R'_{\mathrm{reenc}}$, with S_{reenc} the re-encode share (measured ≈ 0 where probed: the control row of Table 7). The subtraction cancels only the re-encode component of the data share (up to a cross term suppressed by $\sqrt{S_{\mathrm{reenc}}}$, bounded by the control's near-zero reading), so $\|\Delta\bar r\|$ is read net of re-encode; the graph share and the interaction pass through untouched, since the control has nothing to cancel them with.

The drop bounds. Both discarded terms obey exact bounds in the retained shares (Cauchy–Schwarz on the expansion above),

$$|I_e| \;\le\; 2\sqrt{S_e\,\Phi_e}\,, \qquad \big|R'_e\big| \;\le\; 2\sqrt{\big(S_e + \Phi_e + I_e\big)\,\rho_e} \;+\; \rho_e \;+\; O\big(\kappa_\perp^2\kappa_\parallel,\ \kappa_\perp^4\big), \tag{21}$$

with $\rho_e = \|R_e^\perp\|^2/(2\|\bar g_{\mathrm{src}}\|^2)$ the remainder branch's own share. The first says the interaction can never exceed the retained sum ($2\sqrt{S_e\Phi_e} \le S_e + \Phi_e$) and is automatically negligible wherever one share dominates the other, so assumption (ii) carries independent content only near share crossover, which is exactly where Appendix C finds it failing. The second says the first-order part of the remainder enters at half power, one factor of $\sqrt{\rho_e}$ per retained share, so assumption (i) reduces to the branch-level statement that R_e is small against B_e and C_e, plus the small-κ domain for the tail. The control counterpart R'_{reenc} needs no bound of its own: the control's degenerate expansion is itself the measured near-zero row of Table 7, which pins R'_{reenc} jointly with S_{reenc}.

What is not derived. The loading $\|\delta g^\perp\| = a_e\,\|\Delta\bar r(\sigma)\|$ (that the orthogonal gradient mismatch is proportional to the residual mismatch with a σ-independent route gain) is an empirical ingredient (the data-branch factorization at the amplitude level), fit and tested held-out in Appendix C. The geometry above fixes only how a given δg is read into gap units. What *is* derivable from first principles is only a one-sided envelope, $\|\bar J^\top\Delta\bar r\| \le \sigma_{\max}(\bar J)\,\|\Delta\bar r(\sigma)\|$, which would turn the data term into an inequality in the observed mismatch norm given a σ-uniform bound on the mean Jacobian's top singular value. Promoting the envelope to a proportionality with a σ-independent gain requires the mismatch *direction* $\widehat{\Delta\bar r}(\sigma)$ to hold a σ-stationary alignment with $\bar J$'s singular directions (isotropy, for instance, would give gain $\|\bar J\|_F/\sqrt{n}$) while $\bar J$ itself varies with σ through the network input; the image- and route-specific mismatch directions of Appendix B license no such axiom. Hence the loading stays an assumption with a designated probe rather than a theorem.

B Instrument details and secondary reads

Estimator. Per image, arm, and σ-bin, adapter gradients are accumulated over D stratified draws and flattened; cosines are taken between accumulated vectors. Per-bin cosines at D=8 are not comparable *across* bins (the floor drops where $\|g\|$ is small); the gap subtraction against the same-bin floor is the valid read. Split-half reliability over images of each bin-mean curve is mandatory (observed 0.73–0.88 on verdict runs); a per-image variant of the same quantity has split-half reliability ≈ 0 and is not used.

Controls. The re-encoding arm prices the VAE decode–encode round trip that demotion necessarily pays; in pre-debiasing runs its gap served as the validity band (± 0.04; observed $|\cdot| \leq 0.054$ across all bins of the main runs, per-module-group ≤ 0.045 in the split runs), a role superseded by the paired debiased read of §4.1. The redraw floor prices finite-draw noise. Density-weighting the uniform bins by the trainer's σ-density reproduces earlier σ-marginalized measurements from the same instrument family (consistency check).

Debiasing, implementation. Under `--self_floor`, every arm (re-encoding and each demote variant) runs a second independent draw set from a disjoint seed stream, giving per-arm self-cosines alongside the native floor; Eq. 7 is then computed per image and bin, and the map read is the paired excess $\mathrm{gap}_{e,i} = \widehat{\mathrm{gap}}_{e,i} - \widehat{\mathrm{gap}}_{\mathrm{reenc},i}$ with non-finite values (negative self-cosines under the square root at low D) and $|\mathrm{gap}_{e,i}| > 1.5$ trimmed (0–5 of 40 images per cell). Under `--draw_sweep`, draw prefixes are nested ($D = 64$ contains the $D \leq 32$ sets; prefix sums, no extra forwards) and $\mathrm{gap}(D) = \mathrm{gap}_\infty + c/D$ is fit on route means with a bootstrap CI over images. Finite-draw cosines are kernel-path chaotic: twin runs sharing a warm compiler kernel cache agree to $|\Delta \cos| \leq 0.015$, while a run compiled to a different kernel set lands up to ≈ 0.3 away at D=2, so gap/floor/self-floor pairings are not compared across processes, a guarantee the single-run instrument provides by construction; a `--deterministic` mode (bit-exact across twin runs) covers the paired training A/Bs.

Finite-draw bias, measured. Detail behind the magnitudes quoted in §4.1: the c/D fit to the uncorrected 1024→896 endpoint decay gives $c \approx 0.5$. The native floor makes the point sharpest: at the verdict grid's D=8–16 the uncorrected endpoint self-cosine reads 0.79–0.85, a naive read pricing the native gradient's agreement with *itself* at a 0.15–0.2 gap, before the nested sweep ($D = 4 \ldots 64$) extrapolates it to 1.005 (bootstrap 95% CI [0.994, 1.016]).

The U-shaped denominator (behind §4.3). The regimes of the total gradient norm $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ (Fig. 7b): large at high σ, where the model pulls composition out of the prior; small in the mid-σ refinement regime; inflated again at low σ by irreducible ϵ. The renormalization check reads the amplification directly: multiplying each bin-mean gap by $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ flips the anomalous low-σ dip into the monotone-in-σ maximum in every arm of both verdict runs, which is the low-norm amplification behind the interior peak of the measured curves.

Endpoint and x-zero modes. The σ=1 endpoint bin feeds input exactly ϵ (the input-mediated data term vanishes by construction). The x-zero mode zeroes x in input *and* target on every grid, keeping captions and exact demoted latent shapes; with the $(1-\sigma)x$ term absent, low-σ bins are off-manifold and the σ=1 read is primary. In that regime the residual is $\approx -\hat{x}_{\mathrm{prior}}$, so "graph-dominated" includes the model's grid-conditioned prior; the prior-distance probe (§4.3) subsequently dissociated the prior from the floor's route ordering. In the target-strength (α) sweep the interior points $\alpha \in [0.5, 0.75]$ are unreadable by construction and excluded: there the native residual $\alpha x - \hat{x}$ passes near cancellation, the gradient norm dips ($60 \to 33$) and the redraw floor falls to 0.87, so the estimator reads through a small gradient norm: the low-norm amplification above, surfacing along the α axis. The 512 route's sweep anchors are $+0.269 \pm 0.041$ at α=0 vs $+0.337 \pm 0.067$ at α=1, α-flat like the 768 anchors quoted in the main text.

Parallel-landing decomposition of the target term (behind §4.3). The α-flatness has an exact mechanism. Because the objective is quadratic, $\bar{g}$ is affine in α, so the exact target-mediated gradient $t = \bar{g}(1) - \bar{g}(0)$ is computable from the endpoint arms; measured, it is real and large ($\|t_{\mathrm{src}}\|/\|\bar{g}_{\mathrm{src}}\| \approx 2.2$, so $J^\top$ does not annihilate target content), but demotion's change to it lands almost entirely *along* $\hat{g}_{\mathrm{src}}$ ($\kappa_\parallel = -0.75/-1.18/-1.86$ against $\kappa_\perp = 0.09/0.14/0.20$, an 8–9× parallel dominance reproducible across draw sets). The cosine gap is blind to a parallel rescaling (Eq. 18) and the orthogonal part enters only at second order ($\kappa_\perp^2/2 \approx 0.004$–$0.02$), so the α-flat result stands *because* the target term lands parallel, not because it vanishes. In this read the per-image orthogonal loading is ~10× the aggregate's: image-specific directions cancel in the mean, the same motif as the residual-direction read below. For 896 and 768 the floors sit at or below the interior bins' certification resolution, which the endpoint sweeps resolve only through their order-of-magnitude larger draw budget. The statement "the high-σ plateau is the floor" therefore sharpens to: the floor is what the plateau converges to once the data term dies and the estimator can see it.

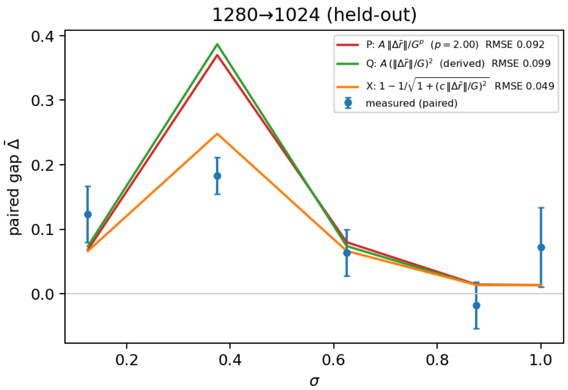


Figure 3: The second held-out route, 1280→1024, under all three functional forms for the data term of Appendix C (fitted power, derived quadratic, exact angular link; legend $G := \|\bar{g}_{\mathrm{src}}(\sigma)\|$), predicted from governors fitted on other routes. This is the one route where the forms visibly separate: the derived quadratic systematically overshoots the mid-σ peak; the exact angular link's saturation removes the overshoot (RMSE 0.049 vs ≈ 0.095), which is why the exact link headlines in Fig. 1.

Direction-resolved mismatch (behind §4.3). The route-uniformity of $\|\Delta\bar{r}(\sigma)\|$ is a property of the *amplitude* only. Saving the per-image mismatch vectors and comparing $\Delta\bar{r}$ directions with a split-half attenuation correction, non-adjacent route pairs are near-orthogonal at low σ (corrected cosine 0.00–0.08 at $\sigma \leq 0.375$) with only a weak shared component at high σ (+0.2–0.3 at $\sigma \geq 0.875$); the top mode of a stacked SVD carries 0.33–0.36 of the energy against the 0.25 rank-4-uniform baseline (a rank-one common mode would give ≈ 1); and cross-image direction consistency is zero at every (route, σ) cell (max +0.019 at split-half reliabilities 0.73–0.99), so the mismatch direction is fully image-specific, refuting a grid-conditional composition prior ("small canvas $\Rightarrow$ portrait framing") as the carrier, at least under full captions. What *is* shared is scale and spectral drift: $\Delta\bar{r}$ energy migrates toward low frequencies as σ rises (low-third share 0.40 $\longrightarrow$ 0.67 by $\sigma = 0.875$), composition-scale but per-image. Eq. 6 only ever consumes the norm, so the account is untouched; what sharpens is the open question, now "why is only the *amplitude* universal when the directions are image- and route-specific." A closed-form candidate for cross-band coupling exists, namely the flow posterior covariance, whose off-diagonal $D_z v^*$ entries are $-\big((1-\sigma)/\sigma^3\big)\,\mathrm{Cov}(x_\omega, x_{\omega'} \mid z)$ (Xing et al., 2026) and are set to zero by the diagonal-Gaussian premise; but its rank-one common-mode version is excluded by the SVD read above.

Held-out validation and refit, sources. The fit consumes the debiased paired curves of the 1024-tier verdict run, the 1280-tier curves (pre-debiasing paired reads; see §6), the route-uniform mismatch norm $\|\Delta\bar{r}(\sigma)\|$ from the residual probe, and each run's own bin-mean gradient norm $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ (gap and norm always from the same process, per the kernel-path rule). The shared-power fit scans $p \in [1, 2]$ jointly across fit routes with per-route weighted least squares; the exact-link fit is a per-route grid with profile-likelihood uncertainties on a_e. Oracle baselines are quadratic-in-σ fits on the held-out data itself (the noise ceiling for a 3-dof curve); the spectral baseline on its committed region is gap = 0 for $\sigma > \sigma_{\mathrm{eq}}$, and its curve-level transport is the bridge of Fig. 5.

The floor law's identifiability. Two anchors and one held-out check cannot identify a functional form, and the cosine gap is the wrong scale to fit one in: it saturates (Eq. 18). Refitting the same anchors in the unsaturated perturbation-energy units $\kappa^2_{\mathrm{eff}} = (1 - F)^{-2} - 1$ gives $\tau \approx 860$ tokens and predicts $F(2160) = +0.096$, equally consistent with the measured $+0.092 \pm 0.012$; the same two anchors pushed through $e^{-\sqrt{n}/\ell}$ or a power law n^{-p} predict +0.085 and +0.075. Every form lands within the union of the measured CI and the prediction's own bootstrap spread, so the exponential is one member of a family our operating points cannot distinguish, and the mechanical reading of τ ($\approx$ 860–1040 tokens across the two unit conventions) as an effective-rank decay length of the early-block gradient operator is a hypothesis, not a law. It is a discriminable one: a spectral-tail account predicts a source-capacity dependence $F \propto e^{-n/\tau} - e^{-n_0/\tau}$ where the absolute-target-capacity governor predicts none (at our operating points the secant correction sits within calibration slack, so the existing anchors do not discriminate); the designated run is a fixed-target, varied-source token ladder, not yet scheduled.

Bootstrap bands and the leave-896-out lane (Fig. 1). The bands are a full-pipeline image bootstrap (B=1000, fixed seed; 981 draws kept): resample images with replacement within each run (N=40 / N=24), re-bin the paired curves, refit the exact-link (a_e, Φ_e) on every fit route, re-derive the ratio governor and floor law, re-predict. $\|\Delta\bar{r}(\sigma)\|$ and the bin-mean $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ are run-level instruments with no per-image decomposition and are held fixed. The replica includes the committed floor law's positive-floor filter ($\Phi_e > 0.005$): in 41% of draws the resampled 1120 floor clears the filter and the law re-anchors on the $(896, 1120)$ legs rather than $(512, 896)$. The bands therefore contain this branch variability, which is the reason §5 deploys the measured map rather than the predicted one. The 896 curve in Fig. 1 is the post-hoc leave-896-out lane under the exact link (c from the ratio-twin 1120 alone = 0.142 vs 0.150 fitted on 896 itself; floor law through 512+1120, whose fragile Φ_{1120} leg enters the log fit clamped positive (36% of bootstrap draws clamp), giving $F(3012) = +0.012$ vs measured +0.019 [+0.010, +0.030]; curve RMSE 0.073).

Interventional B/C ledger, implementation. Routes 1024→{896, 768, 512}, $\sigma \in [0.5, 1.0]$ in four bins plus the σ=1 endpoint bin, D=8 draws per bin, N=24 images, deterministic mode, two independent draw sets per arm. Each arm's draw-summed adapter gradient is stored per bin; the interventional split is $B = \bar{g}_{\mathrm{repromote}} - \bar{g}_{\mathrm{src}}$ (the data intervention at fixed native graph: the demote–repromote arm's input passed through the downscale→upscale→encode pipeline but evaluated on the native grid) and $C = \bar{g}_{\mathrm{demote}} - \bar{g}_{\mathrm{repromote}}$ (the graph intervention at fixed demoted data). The quadratic shares S, F, I and $\rho = \cos(B^{\perp}, C^{\perp})$ use cross-draw-set inner products to cancel shared draw noise; the debiased estimates are not confined to $[-1, 1]$ (hence $\rho = -1.24$ at one low-amplitude bin). The exact counterfactual angles $h(B)$, $h(C)$, $h(B{+}C)$, with $h(X) = 1 - \cos(\bar{g}_{\mathrm{src}}, \bar{g}_{\mathrm{src}} + X)$, are computed on the raw arm means; since $\bar{g}_{\mathrm{dem}} = \bar{g}_{\mathrm{src}} + B + C$ by construction, the realized demotion distance is exactly $h(B{+}C)$, so each account's predicted-to-realized ratio is direct. The no-interference scalar account S+F overpredicts the realized in-window gap 2.4–3.8× at 768 (1.6–5.8× at 896, 1.2–3.0× at 512); the fully additive counterfactual $h(B) + h(C)$ overpredicts 3.5–8×, the realized $h(B{+}C)$ being only ∼20–30% of the additive sum in-window. Including I is what removes the overprediction; the vector account B+C is exact by construction. One unit-honesty note on Table 3: in-window $|B^{\perp}|, |C^{\perp}| \approx 0.5$–$1.0\,\|\bar{g}_{\mathrm{src}}\|$, outside the quadratic truncation's domain, so the truncated ledger S+F+I *under*predicts realized magnitudes ∼3–4× (median ratio 0.24×). The ledger licenses sign, decomposition, and window localization; gap magnitudes are quoted from $h(\cdot)$. Finally, the re-encoding proxy check re-references the data leg against the control arm, $B_{\mathrm{reenc}} = \bar{g}_{\mathrm{repromote}} - \bar{g}_{\mathrm{reenc}}$: $|B^{\perp}_{\mathrm{reenc}}|$ agrees with $|B^{\perp}|$ to ∼4% at the signal-carrying low-σ bins (worst ±23% only where B is already small), so the shared downscale→encode pipeline cost is a minor share of the data intervention. This closes the one open item on the floor ledger's first share, where the re-encoding control (native decode→re-encode) is a *proxy* for the pipeline cost demotion actually pays.

Floor decomposition, detail (behind §4.3). Splitting the probe gradients per module type (15 groups, including separate rows of the fused QKV projections) and per block (28) localizes the floor in *depth*, not module type: early blocks (0–9, peaking at 3–8) carry ∼3× the late-block gap, uniformly across every module type within a block (at $512/\sigma$=1 every type sits in 0.22–0.31; Table 13). The mechanism picture: the first ∼10 blocks build grid-calibrated token statistics, the divergence propagates into every parameter type's gradient in those blocks roughly equally, and deeper blocks inherit a washed-out version. Query and key projections show *zero* excess over value projections, which we initially read as refuting a rotary mechanism. That was an instructive error, since landing-side uniformity cannot localize *origin*: a perturbation originating in the position embedding propagates through the block and lands on all module types. Hence the origin-side positional-interpolation intervention (Table 14; the paired Δs are same-grid contrasts at matched draws, in which the finite-draw bias cancels to first order). The 896 route's small floor sits below that probe's paired resolution and remains undecomposed. A σ-resolved follow-up closes the tempting route this opens (training 1024→768 through interpolated coordinates): with image content in the input the stretched forward is off-manifold, and the interpolated arm is *worse* than the plain demoted arm through $\sigma \in [0.56, 0.81]$ (paired −0.05 to −0.11), winning only at $\sigma \geq 0.94$; the 768 route's data term is in any case fatal on its own (+0.09–0.22 over control across the window). On the non-positional residue, the attention-over-N piece has inference-side prior art: attention-entropy corrections derive length-dependent tem-

Table 3: The interventional B/C ledger behind Appendix C: per (route, σ-bin) orthogonal amplitudes (units of $\|\bar{g}_{\mathrm{src}}\|$), interaction geometry, the cross-set-debiased quadratic shares, and the exact counterfactual angles. In-window S+F+I is sign- and localization-accurate but underpredicts $h(B{+}C)$ (small-perturbation truncation out of domain); magnitude claims in the text use $h(\cdot)$.

route	σ	$\lvert B^{\perp}\rvert$	$\lvert C^{\perp}\rvert$	ρ	S	F	I	$h(B)$	$h(C)$	$h(B{+}C)$
896	0.563	0.55	0.59	−0.95	0.103	0.115	−0.206	0.107	0.206	0.037
	0.688	0.42	0.39	−0.91	0.036	0.059	−0.085	0.083	0.077	0.062
	0.813	0.20	0.23	−0.94	0.016	0.019	−0.033	0.022	0.027	0.010
	0.938	0.10	0.10	−1.24	0.002	0.003	−0.006	0.007	0.006	0.003
	1.000	0.05	0.06	−0.71	0.001	0.001	−0.001	0.002	0.003	0.002
768	0.563	0.57	0.68	−0.93	0.139	0.200	−0.309	0.107	0.346	0.088
	0.688	0.56	0.57	−0.93	0.100	0.136	−0.218	0.106	0.242	0.098
	0.813	0.35	0.42	−0.93	0.049	0.067	−0.105	0.059	0.110	0.035
	0.938	0.14	0.17	−1.02	0.005	0.009	−0.014	0.010	0.033	0.008
	1.000	0.07	0.09	−0.62	0.001	0.004	−0.003	0.003	0.014	0.013
512	0.563	0.60	1.09	−0.72	0.166	0.403	−0.372	0.085	0.854	0.466
	0.688	0.63	0.88	−0.86	0.153	0.332	−0.388	0.107	0.623	0.305
	0.813	0.57	0.72	−0.87	0.152	0.226	−0.323	0.131	0.285	0.126
	0.938	0.25	0.29	−0.96	0.026	0.036	−0.058	0.028	0.201	0.037
	1.000	0.13	0.15	−0.88	0.007	0.010	−0.015	0.007	0.379	0.098

Table 4: **Head to head** on the same measured curves: RMSE in cosine units over all σ-bins. The spectral column is the family transported through our bridge (Fig. 5) and is δ-inert. Our two columns are the derived small-perturbation form (Eq. 6) and the exact angular link (Eq. 18), both with *zero per-route freedom* on held-out routes: amplitudes and floors come from the two governors of §4.3 fitted on $\{1024{\to}896,\ 1024{\to}512,\ 1280{\to}1120\}$. The oracle is a quadratic-in-σ fit on the held-out data itself, the noise ceiling for a 3-dof curve, so beating it on 1024→768 means the prediction is at the data's own resolution.

route	status	spectral	two-term	exact link	oracle
1024→768	held out	0.147	0.093	0.093	0.105
1280→1024	held out	—	0.092	**0.049**	0.056
1024→512	fit route	0.355	0.093	—	—

peratures to preserve the $\log N$ entropy term (Li et al., 2025), and TIDE applies the same dilution correction jointly to resolution and diffusion timestep (Liu et al., 2026), the closest prior work to a σ-gated resolution correction, though on the inference side; our gate is training-side and set by measured gradient safety, not by an entropy invariance.

The floor as a ledger. Combining the designated probes of §4.3, the floor decomposes additively, each share measured by its own intervention:

$$\Phi_e = \underbrace{\text{reenc}}_{\approx 0 \text{ by control}} + \underbrace{\text{target content}}_{\text{lands} \parallel \hat{g}_{\mathrm{src}}\text{: angular share} \approx 0} + \underbrace{\text{RoPE}_e}_{\text{erased by PI at } \sigma=1} + \underbrace{\text{Resid}_e}_{\text{remainder}}$$

with totals 0.02–0.04 (896; below the PI probe's paired resolution, undecomposed), ≈ 0.07 (768; RoPE the large majority, Resid ≈ 0), and ≈ 0.30 (512; RoPE ≈ 0.10, Resid ≈ 0.20).

C Head to head: predicting held-out routes

The probes of §4.3 test each account's *structure*. The remaining test is predictive: fit the reduced account on three routes, then predict two held-out routes from the governors alone. Fit routes: $\{1024{\to}896,\ 1024{\to}512,\ 1280{\to}1120\}$; held out: $\{1024{\to}768,\ 1280{\to}1024\}$. Per fit route the excess curve $\overline{\mathrm{gap}}_e(\sigma)$ is fit with a route amplitude and floor on the measured route-uniform $\|\Delta\bar{r}(\sigma)\|$ and each run's own $\|\bar{g}_{\mathrm{src}}(\sigma)\|$; two governor models ($A$(ratio) interpolating the fitted amplitudes, and an exponential floor law $F(n) = F_0 e^{-n/\tau}$ in target tokens n) then generate the held-out predictions with zero per-route freedom. Pass criteria were pre-registered in the analysis script before the numbers.

Held-out results, and the comparison. All four pre-registered gates fire (Table 4, Fig. 1). (i) Held-out 1024→768: RMSE 0.093, *better than an oracle quadratic fit on the held-out data itself* (0.105), and better than the spectral account both across all bins (0.147) and on the spectral account's own committed region, where it predicts gap 0 (0.164 against 0.096 for ours). (ii) Held-out 1280→1024: RMSE 0.092 under the derived quadratic and 0.049 under the exact link, against oracle 0.056; inside the pre-registered 2× gate either way. (iii) The ratio governor holds at the amplitude level: the two ratio-0.875 routes agree at $z = 0.14$ despite a 1.6× difference in target capacity. (iv) The floor law $F(n) = 0.70\, e^{-n/1041\,\mathrm{tok}}$, fit on the 512 and 896 floors, predicts $F(4825) = +0.007$ for the 1120 grid (fitted +0.002) and $F(2160) = +0.088$ for the held-out 768 grid, against a measured endpoint floor of $+0.092 \pm 0.012$; a post-hoc leave-896-out re-derivation from 512+1120 alone predicts the 896 floor at +0.019 against the measured +0.019 [+0.010, +0.030]. The floor law is thus correct on both floors outside its fit set, though its point estimates ride the draw: a full-pipeline bootstrap puts $F(2160)$ at 68% [+0.042, +0.091], so the licensed read is agreement, not exactness.

The margin in Table 4 has a structural reading, not merely a numerical one. The spectral family's error is not a bad tolerance, since δ is inert, but a missing term and a missing governor: it has nothing to put in the high-σ region where the floor lives, and nothing to distinguish 1280→1024 from a same-ratio route at a quarter of the token count. Our account's held-out accuracy comes from exactly the two ingredients it adds.

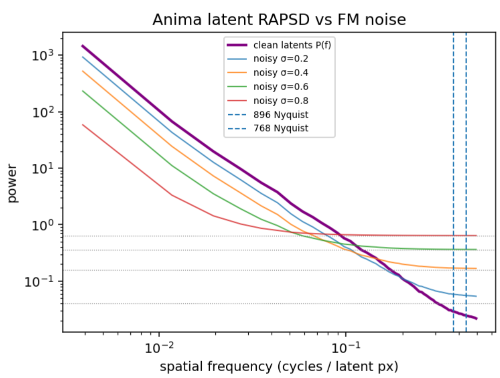
Which functional form, and what the fit does not identify. Refitting the same pipeline under three forms for the data term (a fitted power $A\, m/G^p$ with p shared and free, the derived quadratic, and the exact link) resolves the form question in the geometry's favor twice over. The free exponent's own optimum lands on $p = 2.00$ exactly, recovering the derived quadratic on its own; and the exact link is the best held-out predictor (mean RMSE 0.071 vs 0.093–0.094), with its entire margin coming from 1280→1024, where the quadratic systematically *overshoots* the mid-σ peak and the exact link's saturation removes the overshoot (Appendix Fig. 3). The governors are form-invariant. Two things the fit does *not* establish: the prediction is not within certification resolution anywhere (χ^2/bin 7.6–9.2 held-out under the quadratic), so the licensed claim is shape, magnitude class, and governors at ~0.07–0.09 RMSE; and the floor law's functional form is unidentified at our operating points, as two anchors and one held-out check cannot separate $\{e^{-n/\tau}, e^{-\sqrt{n}/\ell}, n^{-p}\}$, all of which land within the union of the measured CI and the prediction's bootstrap spread (Appendix B).

The reduction's domain: one signed failure, and its mechanism. The 768 route's mid-σ window is the reduction's one structural failure, and its shape is diagnostic. Measured excess sits at zero across $\sigma \in [0.56, 0.94]$, *below* the predicted $\Phi_{768} \approx 0.09$ and outside the bootstrap 95% band (Fig. 1): a positive two-term form cannot produce a gap under its own floor, under any coefficients. In the four-term expansion (Eq. 20) this admits exactly two mechanisms, both pre-registered ahead of the probe: either the projected interaction is negative in the window ($I_{768} < 0$, in which case amplitude matching predicts the window center sits where the two perturbation legs have equal orthogonal magnitude, a testable localization), or the graph share is itself σ-dependent, rewriting assumption (iii) rather than (ii). The designated probe is vector-resolved: a demote–re-promote arm reads the shares per (route, bin) from the interventional split $B = \bar{g}_{\mathrm{repromote}} - \bar{g}_{\mathrm{src}}$, $C = \bar{g}_{\mathrm{demote}} - \bar{g}_{\mathrm{repromote}}$ (Appendix B, Table 3).

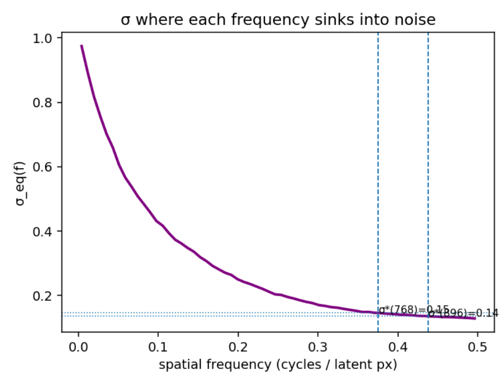
Run, the data picks the first branch with its localization confirmed. $I_{768}(\sigma) < 0$ at every bin (−0.31 at $\sigma = 0.56$, shrinking to −0.014 by 0.94), with the two legs near anti-parallel in-window ($\rho = \cos(B^{\perp}, C^{\perp}) \approx -0.93$), and the predicted window center lands at $\sigma \approx 0.69$ ($|B^{\perp}|/|C^{\perp}| = 0.98$ there), interior to the measured window. Branch (ii) did not fire: $\Phi_{768}(\sigma)$ decreases monotonically to its endpoint value and never drops below it, so the endpoint reads of §4.3 are untouched. The anti-parallel geometry turns out to be universal ($\rho \in [-1.24, -0.62]$ across every route and bin), so what distinguishes routes is amplitude matching, not sign; measured against the exact counterfactual angles, roughly three quarters of the additively-predicted gap is erased by the interference in-window (Appendix B). The window therefore *delimits the reduction's domain*, and the derived four-term form represents the failure rather than excusing it.

Table 5: The spectral account (Eq. 4) instantiated on the measured latent spectrum (P at the 896/768/512 cuts = 0.025/0.029/0.065; RAPSD in Fig. 4): predicted safe boundaries t^* per route as the tolerance δ sweeps its range. δ_{reenc} is the measured re-encode noise floor, a zero-free-parameter anchor fixed by our own pipeline. No row reproduces the measured pattern (last line): the δ that matches the safe route's boundary predicts both failing routes safe by $\sigma = 0.63$, and no δ can produce a floor.

tolerance δ	1024→896	1024→768	1024→512	
$(1 + P_\omega)/2$ (equal power)	0.14	0.15	0.20	classical crossover
0.025	0.50	0.52	0.62	tuned to the measured 896 gate
0.01	0.61	0.63	0.72	default of Xiao et al. (2026)
$\delta_{\text{reenc}} \approx 10^{-5}$	0.98	0.98	0.99	the parameter-free anchor
measured, debiased	≈ 0.5	none certified	unsafe (8/9 bins)	floors .02–.04 / .06–.09 / ≈ .3



(a) Latent RAPSD against the flow-matching noise floor at four noise levels; dashed lines mark the demoted grids' Nyquist cuts.



(b) The equal-power slice $\sigma_{\text{eq}}(f)$ of the spectral family. Predicted σ^*: 0.136 / 0.146 / ~0.20 for demotion to 896 / 768 / 512.

Figure 4: The spectral account, computed from the measured RAPSD and transported to the training question (§3.2): safety would be predicted from $\sigma \approx 0.14$. Against the measurement (Fig. 7) no tolerance reproduces the pattern (Table 5): the equal-power crossover is off by ~3.5×, and the 2× downscale, predicted safe by every tolerance, certifies at no noise level.

D The spectral account: spectrum and instantiation

The tolerance δ of Eq. 4 is the account's one free parameter, and this appendix instantiates the whole family. The choice $\delta = (1 + P_\omega)/2$ recovers the often-quoted equal-power (input-SNR) crossover $\sigma_{\text{eq}} = \sqrt{P_\omega}/(1 + \sqrt{P_\omega})$ as one slice; Xiao et al. (2026) ship an inference-side default $\delta = 0.01$; and our own pipeline fixes a zero-free-parameter anchor δ_{reenc} by construction (below).

Table 5 evaluates Eq. 4 at the demoted grids' Nyquist frequencies $f_{\text{cut}} = \frac{1}{2}\, e/e_0$ on the N=40 mean radially-averaged latent power spectrum (64 radial bins; P at the 896/768/512 cuts = 0.025/0.029/0.065; Fig. 4). Across $\delta \in [0.001, 0.5]$ the mildest–harshest spread remains below 0.13, and the 1024→896 versus 896→768 boundaries (cut frequencies 0.4375 vs 0.4286) separate by no more than 0.01. The parameter-free anchor δ_{reenc} of §4.3 is measured by the same estimator: RAPSD of the re-encode arm's latent error (fresh resize–encode minus cached latent, the gradient probe's own control chain) on the same probe set, interpolated at each route's cut frequency. Measured (N=40, 64 radial bins): error variance 2.7×10^{-5} against latent variance 0.419; $D(f_{\text{cut}}) \approx 1.0 \times 10^{-5}$ at all three cuts (p10–p90 over images within $[1, 2] \times 10^{-5}$; per-band SNR $P(f_{\text{cut}})/D(f_{\text{cut}}) \approx 2.5$–$4.4 \times 10^3$), giving $t^*_{\text{reenc}} = 0.98/0.98/0.99$ for 896/768/512.

The "wrong δ" objection, closed by construction. One might object that the right δ has simply not been chosen: SPD's is an inference-side value selected on a speed–quality Pareto ablation

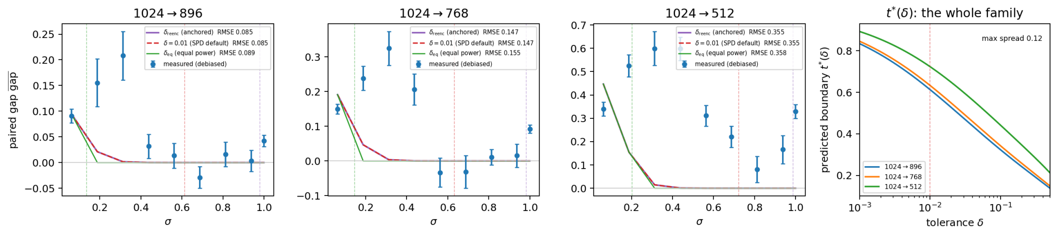


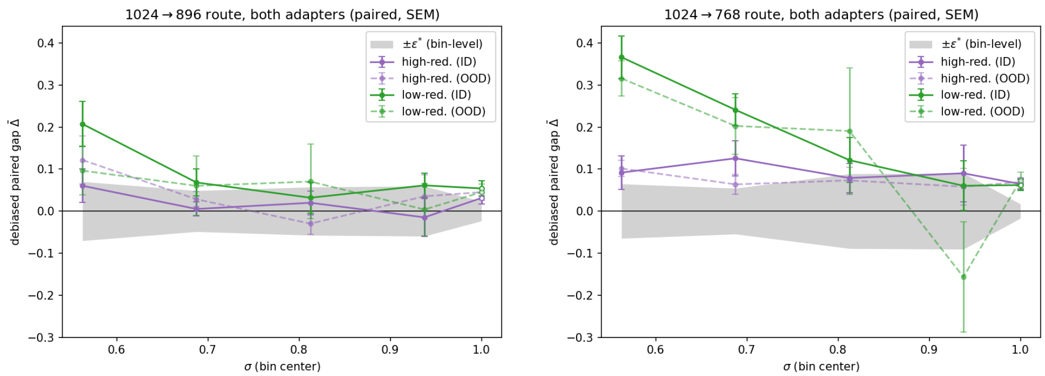
Figure 5: The *whole* tolerance family against the measurement: the detail behind the single red curve of Fig. 1. Left three panels: the measured debiased gap curves against the spectral account transported through our bridge (the diagonal model's destroyed-band mean-residual mismatch on the measured spectrum, through the measured $\|\bar{g}_{\mathrm{src}}(\sigma)\|$ with a single gain calibrated on the one safe route and no floor, which the family cannot express), gated at $t^*(\delta)$ (dashed verticals) for the equal-power slice, SPD's default $\delta = 0.01$, and the measured anchor δ_{reenc}. The transported curve dies by $\sigma \approx 0.35$ on every route, so the two larger gates (0.61–0.72 and 0.98–0.99) act on an already-zero curve and coincide exactly; the equal-power gate (0.14–0.20) is the only one that truncates live curve, and it moves RMSE by at most 0.008. All three therefore land within 0.01 of each other: δ is inert at the curve level. Right: the boundary family $t^*(\delta)$; maximum route spread 0.125 in σ over $\delta \in [10^{-3}, 0.5]$.

over generated images (Xiao et al., 2026), and nothing ties it to gradients. Our setting fixes one *by construction*. The demotion recipe pays a resize–VAE-encode round trip whose latent error has measurable per-band power $D(f)$, expressed in exactly Eq. 4's units, so $\delta_{\mathrm{reenc}}(e) := D(f_{\mathrm{cut}}(e))$ anchors the family with zero free parameters. Measured (above), that anchor is tiny and image-generic, and the boundary moves it forces are measured in Table 5 and Fig. 5: every boundary is pushed to $t^* \approx 0.98$, maximally conservative, and still structurally wrong in both directions at once: it moves all three boundaries together (spread < 0.01), certifies the measured-safe route only above $\sigma \approx 0.98$ where the measurement certifies it from 0.5, and, like every member of the family, predicts the 2× downscale eventually safe when no noise level is. The confrontation of §4.3 therefore does not hinge on the choice of tolerance: the structural failures of §3.2(a–c) are δ-independent at the boundary level, and δ is inert outright at the curve level (Fig. 5).

E Controlled-adapter replication of the verdict map

The one-adapter limitation of §6 pre-registered a controlled 2×2 factorial: two plain-LoRA checkpoints trained under the original probe adapter's frozen recipe on opposite style clusters (the top and bottom 12 artists by median latent redundancy, hereafter the high-redundancy, visually flat, and low-redundancy, heavily textured, clusters), with stem-level membership manifests frozen before training (124 training images each, identical seeds). Each adapter was probed on 48 stems spanning four membership cells (N=12 each): trained-on, held-out images of trained artists, unseen artists of the same cluster, and unseen artists of the opposite cluster, with the cross-cluster stems shared between the two runs so cross-adapter contrasts read paired per stem. Figure 6 repeats the verdict-map recipe per route, overlaying both adapters.

Three reads, stated at the bin-mean level (per-cell tables reproduce from the archived per-image rows). First, the (route, σ) shape replicates on both checkpoints (Fig. 6). Second, the factorial contrasts are null at instrument resolution: the probe-style main effect and the adapter×probe-style interaction (the pre-registered verdict quantity) are both bounded below the resolvable bin-mean effect (raw-gap paired interaction -0.022 ± 0.027 at 896, -0.015 ± 0.059 at 768), and the membership contrast (trained-on versus held-out) does not replicate in sign across the two adapters. The map is therefore adapter- and content-agnostic on this model at our resolution, supporting a calibrate-once-per-model reading. Third, the one quantity that is *not* checkpoint-invariant is the absolute redraw-floor level: in-window floor cosines sit at ≈0.73 for the high-redundancy-cluster adapter versus ≈0.50 for the low-redundancy-cluster adapter, uniformly across all four membership cells. The floor's level is a property of the checkpoint (its gradient stochasticity), while the safety map built on top of it is not.



(a) 1024⟶896 (the certified route).

(b) 1024⟶768.

Figure 6: The Fig. 7a recipe repeated on two controlled adapters trained on opposite style clusters, one panel per route (N=48 stems per adapter, D=8/bin, high-σ window; same paired debiased estimator and trim; open markers the σ=1 endpoint bin; axes shared across panels). Unlike the main-text figures, color encodes the *adapter* (high- vs low-redundancy cluster); solid curves are in-distribution stems, the adapter's own style cluster (n=36: trained-on, held-out, and unseen-artist cells), and dashed curves are out-of-distribution stems from the opposite cluster (n=12). Gray band: bin-level $\pm\varepsilon^*$ per Eq. 8, median of the two runs' full-N SEMs for the panel's route. The map replicates: on the certified route all four adapter×distribution curves reach the band inside the window, on 768 all four sit above it until the top, and within each panel the dashed OOD curves track their solid ID counterparts.

F Per-bin verdict numbers, and the historical record

Table 6 is the numerical form of Fig. 7a, the paper's verdict map, and Table 7 the debiased floor reads that §4.3 scores the graph term on. The remaining tables are the raw-estimator record: retained for continuity and for the reads defended in §4.1 (one-signed bias; paired same-grid contrasts), not cited as debiased evidence. Table 8 is the uncorrected counterpart of the verdict map on the original coarse grid, and its last two rows are the coarse-grid estimator context (Fig. 7b plots the same two curves on the dense verdict grid).

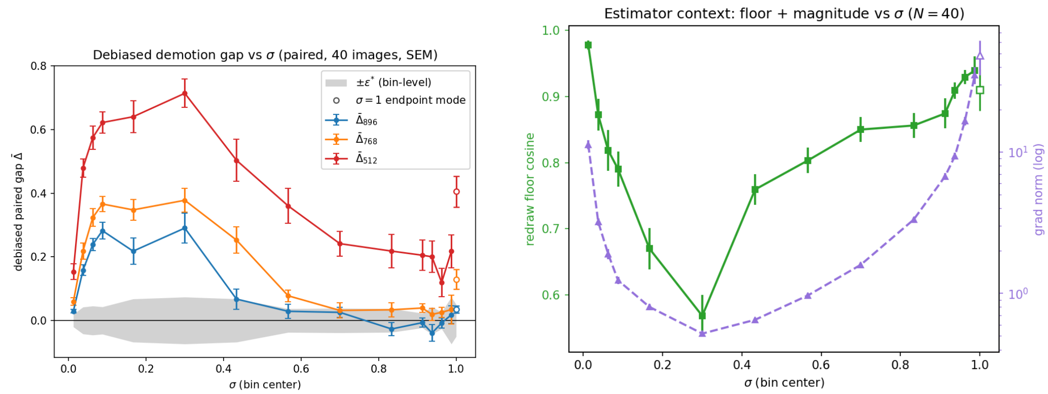


Figure 7: The measurement in detail. (a) Debiased demotion gap per σ-bin, all three routes on one shared axis, paired against the re-encoding control (1024-tier natives, N=40; per-bin values and SEMs in Table 6; gray band: bin-level $\pm\varepsilon^*$ (§4.1); open markers: the σ=1 endpoint bin): safety is a per-route boundary. The 896 route enters the band at $\sigma \approx 0.5$, 768 falls only to a low-excess shelf (+0.02–0.04 over $0.65 < \sigma < 0.95$, short of the strict margin), and 512 never falls inside the band (closest approach $+0.12 \pm 0.05$ at $\sigma = 0.96$); all three boundaries stand against the spectral family's prediction of $\sigma \approx 0.14$–0.20. (b) The estimator context those curves are read through: the redraw-floor cosine and the U-shaped gradient norm $\|\bar{g}_{\mathrm{src}}(\sigma)\|$.

Table 6: **Debiased verdict map** (plotted in Fig. 7a): paired per-image debiased excess $\overline{\text{gap}}$ (SEM) against the re-encoding control, 1024-tier natives (N=40, D=12/bin, deterministic kernels, on a segmented grid dense below $\sigma = 0.1$ and above 0.9: 14 interior bins plus a σ=1 endpoint bin; trimmed per §4.1, retaining 36–40 images per cell). Bin-level ε^* at this (N, D) is 0.02–0.07. A reconstruction check (the realized excess rebuilt from per-arm means) flags the three $\sigma \leq 0.06$ bins as partly estimator inflation, so the raw-paired excess is the primary read there (896: +.037/+.118/+.221; 768: +.065/+.202/+.275; 512: +.172/+.406/+.497), up to 0.08 smaller with the same verdicts.

σ bin center	1024→896	1024→768	1024→512
0.013	+.029 (.006)	+.059 (.012)	+.154 (.025)
0.038	+.158 (.016)	+.219 (.026)	+.480 (.029)
0.063	+.239 (.019)	+.323 (.028)	+.575 (.037)
0.088	+.282 (.026)	+.366 (.024)	+.622 (.033)
0.167	+.218 (.041)	+.348 (.033)	+.640 (.050)
0.300	+.291 (.047)	+.378 (.038)	+.714 (.045)
0.433	+.067 (.032)	+.253 (.041)	+.504 (.065)
0.567	+.029 (.023)	+.078 (.019)	+.360 (.055)
0.700	+.026 (.016)	+.032 (.023)	+.242 (.039)
0.833	−.027 (.020)	+.034 (.023)	+.218 (.053)
0.913	−.007 (.014)	+.039 (.013)	+.205 (.047)
0.938	−.039 (.026)	+.019 (.019)	+.200 (.049)
0.963	−.008 (.015)	+.026 (.016)	+.120 (.045)
0.988	+.018 (.028)	+.034 (.045)	+.217 (.052)
1.0 (endpoint)	+.034 (.011)	+.130 (.030)	+.406 (.049)

Table 7: **The floor, debiased** (scored in §4.3). **Endpoint**: at σ=1 the input is exactly ϵ on every grid, so the input-mediated data term vanishes by construction. **x-zero**: the image is removed from input *and* target; any surviving gap is pure graph-shape sensitivity. Debiased draw-limit values $\widehat{\text{gap}}_\infty$ [bootstrap 95% CI over images] from nested draw sweeps (endpoint: N=12, $D = 4 \ldots 64$; x-zero: N=40, $D = 4 \ldots 32$). The two columns agree within CI at every route; our pre-registered kill switch (all endpoint gaps ≈ 0) does not fire; and no member of the spectral family can produce a nonzero entry here.

route	endpoint $\widehat{\text{gap}}_\infty$ [95% CI]	x-zero $\widehat{\text{gap}}_\infty$ [95% CI]
native self-floor (cosine)	1.005 [0.994, 1.016]	—
re-encoding control	−0.003 [−0.017, +0.008]	—
1024→896	+0.019 [+0.010, +0.030]	+0.034 [+0.017, +0.058]
1024→768	+0.056 [+0.043, +0.071]	+0.074 [+0.053, +0.094]
1024→512	+0.304 [+0.197, +0.424]	+0.283 [+0.232, +0.332]

Table 8: **Raw estimator** (historical record; superseded by Table 6): demotion gap by σ-bin, 1024-tier natives (N=40, bin-mean SEM ~0.02, re-encoding control $|\text{gap}_{\text{reenc}}| \le 0.054$ everywhere, split-half reliability 0.73–0.83). Bold = within the re-encoding band. Raw values understate mid-σ gaps (the floor attenuates the read) and overstate endpoint floors (single-estimate variance bias); the debiased map corrects both.

σ bin center	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
gap_{896} (ratio 0.875)	.110	.162	.148	.137	**.048**	**.030**	**.030**	.053
gap_{768} (ratio 0.75)	.144	.216	.208	.223	.164	.163	.115	.063
gap_{512} (ratio 0.5)	.348	.410	.355	.469	.430	.391	.289	.296
$\cos_{\text{floor}}$	.84	.65	.51	.65	.70	.77	.80	.83
$\|g\|$ (native)	2.6	0.5	0.4	0.5	0.7	1.1	2.0	9.3

Table 9: Raw endpoint reads (D=16, N=40) beside their debiased draw-limit values (Table 7), the debiasing's revision record: the 768 floor halves, the 896 floor surfaces.

route	raw endpoint (D=16)	endpoint $\widehat{\text{gap}}_\infty$ [95% CI]
native self-floor (cosine)	0.85	1.005 [0.994, 1.016]
re-encoding control	−0.038	−0.003 [−0.017, +0.008]
1024→896	−0.009	+0.019 [+0.010, +0.030]
1024→768	+0.127	+0.056 [+0.043, +0.071]
1024→512	+0.326	+0.304 [+0.197, +0.424]

Table 10: Iso-severity probe (**raw estimator**): gap by σ-bin on the 1280-tier cache (N=24, 4 draws/bin) against the corpus 1024→896 curve. The ratio-matched routes coincide within ~1.4 combined SEM at every bin despite a 1.6× difference in absolute target size; the same-native harsher-ratio route separates. *896's canonical 16-draw endpoint is −0.009 raw, +0.019 debiased (Table 7).

σ	0.125	0.375	0.625	0.875	1.0
re-encoding control	+.047	−.005	+.048	+.020	−.012
1280→1120 (ratio .875)	.129	.110	.054	−.012	−.007
1280→1024 (ratio .80)	.170	.178	.111	.002	.061
1024→896 (ratio .875)	.132	.076	.077	.049	.100*

Table 11: Ratio-transfer run (**raw estimator**): 896-tier natives (N=40), pre-registered bar "gap_{768} within the re-encoding band at $\sigma \ge 0.5$": **fails** (residual 0.06–0.12, ~2× the 1024→896 plateau; ~half of it is expected estimator bias, §4.1; the separation is independently confirmed by the debiased same-grid floors, §4.3).

σ bin	0.06	0.19	0.31	0.44	0.56	0.69	0.81	0.94
896→768	.114	.209	.168	.143	.124	.056	.092	.061
896→512	.318	.372	.308	.340	.320	.280	.329	.217

Table 12: Route-uniformity (in amplitude) of the mean prediction residual: cross-grid excess of $\bar{r} = \mathbb{E}_\epsilon [\hat{v} - (\epsilon - x)]$ over split-half floors (relative L_2; same-grid re-encoding control ≤ 0.02 everywhere). The three main routes coincide within $\sim \pm 0.02$ at every σ while their gradient floors span 0 to 0.3. This is a norm read; the direction-resolved follow-up (Appendix B) shows the underlying mismatch directions are image- and route-specific.

σ	0.125	0.375	0.625	0.875	1.0
1280→1024	.885	.825	.715	.465	.360
1024→896	.862	.808	.706	.459	.378
896→768	.860	.814	.715	.466	.393
768→512	.897	.872	.767	.508	.435

Table 13: Depth localization of the floor (**raw estimator**): mean per-block gap at σ=1, early = blocks 0–9, late = 14–27, for the endpoint (ep) and x-zero (xz) probes. Early blocks carry ~3× the late-block gap. The apparent content share (ep − xz) is a late-block minority effect in raw units; the debiased draw-limit comparison finds ep = xz within CI at the whole-adapter level (Table 7). Within a block every module type sits in 0.22–0.31 at $512/\sigma$=1, including the query and key rows of the fused QKV projection: landing-side uniformity, which is why the positional share needed an origin-side intervention to find.

route	ep early	ep late	xz early	xz late
1024→512	.357	.223	.351	.125
1024→768	.164	.085	.121	.033
1024→896	.023	.015	.044	.012

Table 14: Positional-interpolation intervention at the σ=1 endpoint (N=40, 16 draws; **raw** arm values; the paired Δ column is a same-grid contrast in which the finite-draw bias cancels to first order): exact phase-geometry alignment erases the mild-route floor and ~30% of the harsh-route floor.

route	plain (SEM)	PI-aligned (SEM)	paired Δ (SEM)	improved
1024→896 (control)	−0.021 (.041)	−0.040 (.040)	—	—
1024→768	+0.080 (.048)	−0.001 (.039)	+0.081 (.031)	78%
1024→512	+0.320 (.058)	+0.224 (.056)	+0.096 (.039)	70%